AI-Driven Model for Online Payment Fraud Detection

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Preface

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Student's Declaration

I declare that this written submission represents my own ideas in my own words, carried out under the supervision of Prof Dr. Musarat Karim, Department of Information Technology, Faculty of Computing, The Islamia University of Bahawalpur, Pakistan. Where others' ideas or words have been included, I have adequately cited and referenced the original sources. I declare that I have properly and accurately acknowledged all sources used in the production of this thesis. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented, fabricated, or falsified any idea, data, fact, or source in my submission. At this moment, I declare that the substance of this submission has neither been submitted elsewhere nor is being concurrently submitted for any other degree.

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F24BINFT3E02026

Supervisor's Declaration

It is hereby certified that the work presented by Faheem Abbas in this thesis titled "AI-Driven Model for Online Payment Fraud Detection" is based on the results of the research study conducted by the candidate under my supervision. He has fulfilled all the requirements and is qualified to submit this thesis in partial fulfilment for the degree of Master of Science (MS) in Information Technology, Department of Information Technology, Faculty of Computing, The Islamia University of Bahawalpur.

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Certificate

I certify that Mr. Faheem Abbas s/o Nazar Hussain has complied with the Regulations for Higher Degrees in this thesis. I also certify that the research work embodied in this thesis entitled "AI-Driven Model for Online Payment Fraud Detection" has been carried out by Mr. Faheem Abbas under my supervision and is worth presenting to The Islamia University of Bahawalpur, Pakistan for the award of the degree of Master of Science.

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Dedication

I want to dedicate this dissertation to my late mother, Kaneez Rubab, whose prayers for my success have always stayed with me and continue to guide me, and who I know still prays for me from beyond, from her resting place. To my father, who has been my rock and my warrior my entire life and who always prays for me whenever he offers his prayers, how he keeps everything in check is beyond my comprehension. And to my elder brother, who has been a tree of shade for me, offering shelter, comfort, and quiet strength whenever I needed it most.

“My Lord, have mercy upon them as they brought me up [when I was] small”

(Quran 17:24)

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Abstract

Payment fraud on the Internet is a massive global problem and the traditional rule based approach is inadequate to the size and variety of card not present (CNP) fraud. This thesis for the first time gives a systematic comparison of AI based fraud detection methods, filling the gaps found in the existing literature on limited cross paradigm comparison, overall inference latency, and systematic hybrid ensemble of AI based methods.

The models tested were 5 models: Random Forest, XGBoost, SVM, Isolation Forest and Autoencoder as well as a new Hybrid Ensemble of XGBoost, LightGBM and Random Forest, that were tested on 2 datasets: European Credit Card Fraud and IEEE CIS with 6 metrics: Precision, Recall, F1 Score, AUC ROC, False Positive Rate and Inference Time employing stratified 5 fold cross validation and class weighting for imbalance. The top supervised models were identified and key fraud indicating features using SHAP analysis.

The findings reveal that supervised ensembles gain higher performance compared with unsupervised models on both data sets and Hybrid Ensemble has the highest overall AUC ROC and F1 Score and can be implemented in real time. The performance on the more complex IEEE CIS dataset decreased in the experiments, indicating a limited degree of transferability of the approach between environments. V12, V14 and V4 consistently ranked as top predictors on SHAP for European and C13, tr_hour_z and C1 for IEEE CIS.

The Hybrid ensemble is proposed to be used in combination with another model namely Isolation Forest and Autoencoder for anomaly detection in three-layer detection architecture. Notable contributions include the cross paradigm evaluation, the novel Hybrid Ensemble, treating latency as a core metric, SHAP based interpretability verified by Friedman testing. Restrictions on the age of dataset and generalisability are highlighted and future work on integrating ANN and GNN and federated learning is outlined; thereby this research can serve as a stepping stone towards its adoption in the digital financial sector in Pakistan.

List of Tables

Abbreviation	Full Form
AI	Artificial Intelligence
ANN	Artificial Neural Network
ANOVA	Analysis of Variance
ATM	Automated Teller Machine
ATO	Account Takeover
AUC	Area Under the Curve
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
CC	Credit Card
CNN	Convolutional Neural Network
CNP	Card Not Present
CSV	Comma Separated Values
DICE	Diverse Counterfactual Explanations
DS	Data Science
DSS	Decision Support System
EAI	Explainable Artificial Intelligence
EFB	Exclusive Feature Bundling
FIs	Financial Institutions
FL	Federated Learning
FN	False Negative
FP	False Positive
FPR	False Positive Rate
GAN	Generative Adversarial Network
GBM	Gradient Boosting Machine
GNN	Graph Neural Network
GOSS	Gradient based One Side Sampling
ID	Identity / Identifier
IEEE-CIS	Institute of Electrical and Electronics Engineers Computational Intelligence Society

IP	Internet Protocol
KYC	Know Your Customer
LIME	Local Interpretable Model agnostic Explanations
LSTM	Long Short Term Memory
MSE	Mean Squared Error
PCA	Principal Component Analysis
PCI	Payment Card Industry
PSD2	Payment Services Directive 2
ROC	Receiver Operating Characteristic
RQ	Research Question
SBP	State Bank of Pakistan
SHAP	SHapley Additive exPlanations
SIM	Subscriber Identity Module
SME	Small and Medium Enterprise
SMOTE	Synthetic Minority Oversampling Technique
SVM	Support Vector Machine
TN	True Negative
TP	True Positive
USD	United States Dollar
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting

Chapter 1:

Introduction

1.1 Background and Context

1.1.1 The Digital Payment Revolution

The last 20 years of the 21st century has seen a fundamental and pervasive shift in global commerce, the digitalization, networkization, and virtualization of financial transactions from physical, paper-based, and in-person settings to digital, networked, and remote ones, and it has been quick, rapid, and is now largely irreversible. This change hasn't happened overnight or over a single decade, but rather over the past ten years or more, in which a combination of technological, economic, social and regulatory developments have all contributed to redefining the core of any economy: the exchange of value. Billions of individuals without reliable access to formal financial infrastructure were enabled to access payment capabilities with the proliferation of smartphones and affordable mobile internet access services (Du et al., 2023). As secure communications (Transport Layer Security, tokenisation and end-to-end encryption) matured, consumers were confident enough to entrust their financial details with digital channels. Banking and payment card networks, and at the national level, banking and payments systems were expanded into markets and demographic groups where they were not available.

Global players such as Amazon, regional marketplace players and hyperlocal delivery companies paved the way for platform-based e-commerce to emerge, turning any e-commerce purchase into an online-only transaction. These changes became structural and were reinforced by a swift and deep change in consumer behaviour from cash to digital forms, which have since become the norm and are now a much faster and more convenient way for consumers to settle their affairs (Esenogho et al., 2022).

The outcome has been a global digital payments system that is unprecedented in scale, complexity and economic impact. E-commerce accounted for $5.58 trillion of sales in 2023, and is expected to take up 19.9% of global sales in 2024 and 20.5% in 2025. The figures are not just a statistical achievement, but also a structural change in the global economy: enterprises that traditionally operated from premises close to customers are now operating across continents from a single warehouse; consumers transact with businesses they do not have to meet face-to-face and never get to hold, in currencies they neither hold nor have ever seen, and from time zones half a world away; and entire commercial sectors, from streaming media and software-as-a-service to online education or digital content creation and virtual healthcare, exist entirely in the digital payment environment and could not function without it.

This ecosystem is supported by infrastructure that is correspondingly complex, sophisticated and interdependent. Each individual transaction is processed by each of these players, using a set of standardised protocols and settlement processes that are expected to manage high transaction volume, speed and reliability. In the more advanced digital payment space, the speed of the payments system itself is astounding – and it takes less than two seconds for a payment to be submitted, fraud checked, authorised, and the merchant notified, with the decision on whether or not to approve the payment taking effect within milliseconds (Sinha, 2024). The speed requirement is not just a practical consideration for the user but a critical operational constraint that influences various components of the fraud detection system's design, implementation, and assessment.

As the volume has increased, so has the variety of payment channels. In addition to the classic credit and debit card payment options, today's digital commerce includes mobile wallets, buy-now-pay-later financing solutions, payment options that bring cryptocurrency close to the user, peer-to-peer transfer platforms, and even embedded payment within social and messaging apps. Every new channel means new types of transactions, new authentication issues and new vectors for fraudsters. The most quickly accelerating new payment methods are mobile payments, which are now accepted by 52% of merchants, real-time payments with 42%, and buy-now-pay-later methods, which are accepted by 32% of merchants – and all of these are gaining a foothold in the last year alone. With this diversification of the payment landscape comes the need for fraud detection systems to support multiple payment transactions, each with its own set of statistical signatures, velocity patterns and risk profiles; a challenge that rule-based systems are not well suited to meet (Zhao et al., 2022).

1.1.2 The Pakistani Digital Payment Context

In developed economies of North America, Europe and East Asia. At a time when digital payments are transforming global commerce, Pakistan, where a significant portion of the population is young, digitally savvy and connected to mobile devices, is experiencing a digital payment revolution, of a remarkable pace, scale and impact. A high degree of smartphone penetration, low prices of mobile data, a conducive regulatory framework, and substantial investments in digital financial infrastructure have delivered some of the fastest adoption rates of digital payments in any major emerging market.

The global e-commerce market will touch 20.5% in 2025 and the overall trend is similar in Pakistan, as some major ecommerce platforms like Daraz, Foodpanda and a growing number of vertical ecommerce companies in fashion, electronics, grocery, healthcare etc. are growing rapidly. The instant payment system Raast, developed by the State Bank of Pakistan, a government-backed real-time payment system, has significantly simplified the process of digital payments, made it easier to access, and brought convenience to those who rely on cash for their payments. In total, tens of millions of people have registered on mobile payment platforms such as JazzCash and EasyPaisa, many of whom are using formal payment services for the first time via these channels.

This digitalisation has created significant economic and social benefits such as financial inclusion, lower transaction costs, more efficient supply chains and a wider market access for SME's who can now tap into a broader geographic market. It has also driven a proportionately fast increase in fraudulent activity that accompanies growing popularization of any large digital economy. SBP's heightened regulatory focus on digital fraud prevention, such as the issuance of guidelines to commercial banks and microfinance institutions mandated the implementation of real-time digital fraud prevention policies, is a testament to the realisation that the development of Pakistan's digital payments market will need to be supported by an investment in the security infrastructure that will protect it. Social engineering attacks targeting gaps in digital literacy, SIM swap fraud, identity theft, unauthorised account access, and credential phishing targeting mobile banking users are the most common types of fraud in the Pakistani context that are well suited to be detected by AI-driven detection systems (Mohbey et al., 2022). This research, apart from its relevance to the global academic research in payment fraud detection, is also situated within the local context of the rapidly growing electronic financial system in Pakistan as it pertains to the immediate security concerns being faced in the country.

1.1.3 The Global Fraud Crisis

As ecommerce has expanded, so has the amount of fraud going on online, and at an increasing pace. These two trends are interrelated: any new digital payment channel, new type of online merchant, and new customer segment who embrace digital transactions creates a new attack surface for criminals. The global financial risk of ecommerce fraud is predicted to more than double, rising from over USD 40 billion in 2024 to over USD 100 billion by 2029, as ecommerce transactions continue to grow worldwide. Financial institution fraud losses will grow by more than 150% to over USD 55.3 billion by 2030, from less than USD 25 billion in 2025 (Khalid et al., 2023).

The total online payment fraud loss to merchants will reach $362 billion globally from 2023 to 2028, and amount to $91 billion in 2028. These are not ephemeral numbers but losses suffered by businesses and consumers – and by the banks that handle their payments – and the losses result from such charges as fraudulent transactions, goods shipped to a false address that the merchant never receives, and the ongoing expense of fraud operation, chargeback processing and fraud prevention technology that everyone in the payment ecosystem pays.

The total amount of money lost to credit card fraud in 2024 is estimated to be as high as $37 billion, up 19% from the previous year, with many cases resulting from card-not-present (CNP) transactions (Chavakula et al., 2025). The ability to verify transactions in a remote payment environment poses a challenge in e-commerce credit card fraud, with nearly 80% of all e-commerce credit card fraud losses caused by CNP fraud. In the meantime, ecommerce businesses

spend $48 billion annually on fraud, and every $1 spent on fraud equals $4.61 in losses to US merchants.

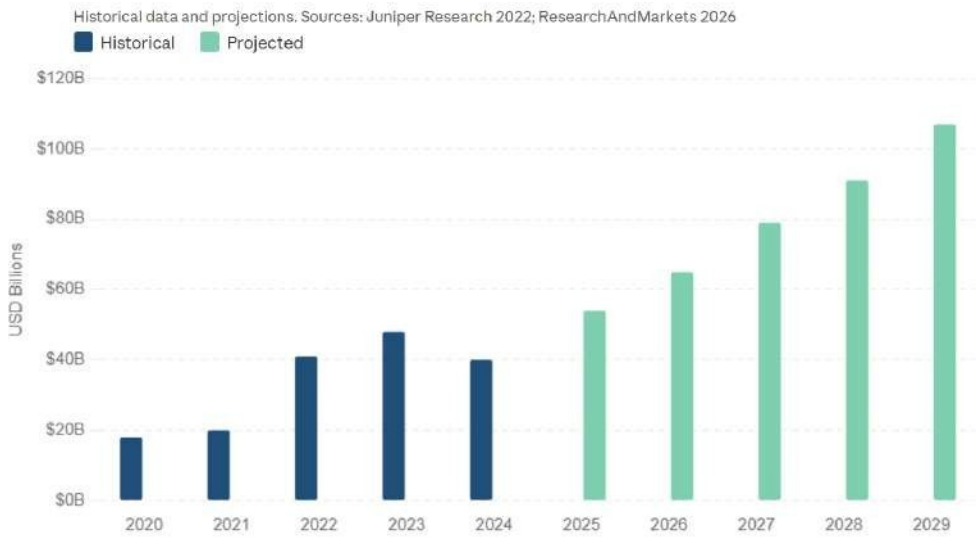


Figure 1.1 Global e-commerce fraud losses 2020–2029

Online payment fraud has a broad impact on the payment economy. When the cards are stolen, the cardholder is directly affected by the fraudulent use and the stress of having to deal with the transaction dispute. Merchants have to bear chargeback fees, investigation charges, and forfeiture of goods already shipped. The burden of fraud reimbursement, customer service and more and more advanced prevention technologies fall on financial institutions. The wider digital economy sees a loss in consumer confidence in online payment systems, resulting in a decrease in the uptake of ecommerce and a failure to reap the potential efficiencies provided by these online payments. There is no one in the payment ecosystem who isn't affected by fraud costs – making investment in effective fraud detection an enterprise-wide interest.

1.1.4 The Inadequacy of Traditional Detection Approaches

The typical reaction to the fraud risk by financial institutions and payment processors has always been the implementation of rule-based detection systems, which are sets of manually-defined logical conditions and threshold values that can automatically identify or reject transactions that include known fraud indicators. A rule-based system could say no to transactions from an unknown device if the amount is greater than a certain threshold, it could reject any two or more transactions in a short period of time using a single device, or it could prevent transactions from locations that didn't match the cardholder's purchases (Ali et al., 2025).

Fraud prevention systems have been and will remain the foundation of fraud prevention for decades and are still valuable as a first line of defence for the most obvious and well-established fraud patterns. The problem with such a rule-based approach has grown increasingly problematic as the sophistication, scale and adaptability of fraud has increased, and as payments have been made at a volume that makes the cost of high false-positive rates too high.

The key problem with rule based systems is that they are static. They are only able to detect fraud patterns that the human analysts have beforehand expected and written down in the form of rules. Every new attack vector that the fraudsters can develop is not visible to the detection system until it has been studied, documented and translated into rules, which takes weeks and/or months, where the new attack vector is free to be exploited (Lee et al., 2025). Sophisticated fraud operations are those that research and test the limits of what is known and understood about the detection rules, and fine-tune their attack patterns to avoid detection. This cat and mouse game between the rule creators and fraudsters is naturally asymmetric: Fraudsters can make changes in hours, and the creators can make changes in weeks.

The second severe restriction is the high false positive rates. For large volume payments, any false positive rate of 0.1% is hundreds of false alarms per hour, each of which represents a lost real payment, a real customer who was unable to make a payment because his/her transaction happened to trigger an incident in a fraud rule. These false alarms often lead to complaints from customers, the need for manual verification by fraud analysts, and ultimately the loss of customer confidence in the system's reliability and reaction. In e-commerce markets where customer experience is a major differentiator, having too many false positives is a cost that can be measured to merchants and financial institutions who use them (Sharma et al., 2022).

1.1.5 The Promise of Artificial Intelligence

Artificial Intelligence and Machine Learning provide a new paradigm for fraud detection that is adaptive instead of static, data based instead of rule based, and can run at the speed and scale of digital payment environments without losing the subtle and contextual insights needed for effective fraud detection.

Unlike having to identify fraud rules upfront by human experts, AI models can be trained with historical data to learn the statistical patterns that are characteristic of fraudulent and legitimate transactions. No human analyst could possibly describe all of the subtle combinations of when, how much, what kind of merchant, what type of device and what sort of account history that together create a good signal of fraudulent intent in a transaction, and that's what this learning process captures. When trained, AI models can then use these patterns for new transactions in mere milliseconds, issuing a risk score that reflects the possibility of fraud and that can inform real-time decisions on whether to approve, refuse, or question a particular transaction (Sarna et al., 2025).

One of the most advantageous properties of AI models is their versatility. Fraud patterns change over time, as new attack vectors arise, as seasonal fraud surges, and as the transaction population changes AI models can be trained on new data to learn new patterns, keeping them effective – static rule-based systems simply can't do that. This flexibility turns fraud detection into a learning process, instead of a rule-updating process, and continually evolves with new data.

At the same time, unsupervised AI methods are particularly well suited to detect fraudulent transactions that do not match any of the observed fraud patterns, like anomaly detection algorithms such as Isolation Forest and Autoencoders. These do not depend on a set of marked

examples of fraud, but rather they learn what "normal" and "legitimate" transaction behaviour is, and flag transactions that are significantly different. This is especially useful for identifying emerging fraud patterns, before enough labeled samples have been gathered for supervised classifiers to learn the pattern (Palivela et al., 2024).

Even with this assurance, AI-powered payment fraud detection isn't without its hurdles. Fraud data sets often are highly imbalanced, with a minority of transactions being fraudulent and a majority legitimate, which can lead to systemic biases when training models on such data. Even AI-powered systems have a sensitivity/false positive trade-off that must be designed and evaluated. Interpretability of AI model decisions, especially for regulatory compliance and for the trust of fraud analysts who have to take action based on model outputs, is not automatic and investment in analysis is required. In real-time payment processing, the inference speed demands of real-time applications set limits that some AI architectures aren't able to meet.

The motivations for this thesis are rooted in the belief that these problems are not insurmountable; that with careful experimental design, effective handling of class imbalance issues, the use of multi-metric evaluation frameworks which explicitly account for the false-positive rate and inference latency, and a systematic interpretability analysis, AI based fraud detection models can be made both highly accurate and practically deployable. The research is designed to bring the evidence in hand to make that conviction actionable (Islam et al., 2023).

1.2 Problem Statement

Payment fraud is one of the most pressing, expensive and technically advanced problems faced by the global digital economy. Over the last 20 years or so there has been a great deal of investment in fraud prevention technologies, and big advances in computing power, data availability, and algorithmic sophistication – and yet fraud continues to increase in absolute dollar value, adapt to new detection environments and exploit the structural vulnerabilities in remote, card not present transaction environments. The problem isn't that no technology exists the machine learning tools that can be used to construct well-functioning fraud detection systems are well-understood, widely available and freely accessible. The issue is that they are very complex and cannot be effectively deployed due to several concurrent challenges with different and even sometimes conflicting demands on the system for detection (Misra, 2023).

Challenge 1 The Rigidity of Rule Based Systems: The financial sector has the most widely used fraud detection systems, and these systems still rely heavily on the underlying logic of rules. The thresholds and logic used in these systems are hard-coded into the system and are only capable of detecting known patterns of fraud which have been designed into the system, and are not capable of adapting to new patterns of fraud without manual effort. Fraudsters know about this blind spot and frequently try to test the boundaries of their fraud detection and identify what is the maximum amount that they can get away with without being detected, and then they'll try to work their way up to that limit. It's an ongoing arms race, in which the defenders have to catch up with attackers,

and since the defenders have to write the rules of their lives — which is a human process — they are always two steps behind. The loss due to this asymmetry is valued at billions of dollars in fraud losses that will happen in the gap between the occurrence of new attack patterns, and the deployment of new rules of detection.

Challenge 2 The extreme class imbalance: In real-life payment transaction datasets, the number of fraudulent transactions is usually less than 0.2% from total recorded transactions. This is a very imbalanced class distribution that poses basic difficulties in the training of machine learning models. A model that correctly classifies all individual examples as legitimate, never detecting any fraud, would have an accuracy of more than 99.8% on this example dataset – and this is a perfect failure at the actual task of fraud detection. Without explicit attempt to correct for class imbalance, models are systematically prone to developing this bias, creating a system that may seem very accurate when assessing, but very limited in its practical fraud detection application when deployed. A key methodological hurdle in fraud detection AI studies is addressing the issue of class imbalance without adding artificial patterns that decrease model generalisability.

Challenge 3 The False Positive Dilemma: There are two competing goals in fraud detection: detection of true fraud is important (high recall); but, the number of "false" detections of legitimate transactions is also important (low false positive rate). Each threshold that is set in a fraud detection system represents a compromise between these goals. The higher the sensitivity, the more transactions that are flagged incorrectly, meaning that a higher rate of fraud is also likely to be detected. Any reduction in the number of false positives will inevitably be at the cost of a reduction in the number of true frauds caught. It is not possible to remove this trade-off, only manage it. Past studies often concentrate on one side of the trade-off and ignore the other, with many studies optimizing for detection measures, but failing to report false positive rates, making it difficult to evaluate the implications of their findings for customer experience.

Challenge 4 Limited Cross-Paradigm Comparative Evidence: The fraud detection research literature exists in two distinct and mostly independent communities—one supervised, working on the classification performance of labelled data, and one unsupervised, working on the unsupervised detection of unusual patterns. Comparative studies are few, where both paradigms are applied to the same data set and under the same experimental conditions, and provide no conclusive evidence for the superiority of supervised versus unsupervised methods and for how they should be layered within a detection architecture.

Challenge 5 Failure to consider Operational Constraints: Most academic fraud detection research is based on classification performance measures while not reporting or measuring any of the other operational constraints such as computational inference time, training complexity and interpretability of model decisions for fraud analysts. A lack of linkage between academic evaluation and the operational system's deployment needs makes it difficult to easily apply the results of academic research to design decisions in production systems, as the benefits of the

model's performance are not relevant if it cannot be deployed for processing in real time or it cannot provide the explanations that are necessary for regulatory compliance.

The present research directly addresses all of the above challenges through a thorough, experimentally sound multi-model comparison across supervised and unsupervised AI paradigms that covers all operationally-relevant performance metrics such as inference latency and in the process introduces systematic SHAP interpretability analysis in order to render model decisions transparent and actionable to practitioners.

1.3 Research Questions

This study is directed by four primary research questions that together address the theoretical, technical, and practical dimensions of AI-driven online payment fraud detection:

RQ1: What can the published literature in 2022 and later tell you about the current state of the art of AI-powered online payment fraud detection and what are the specific gaps in the research?

RQ2: Which preprocessing techniques (Handling missing values, Feature scaling, Train Test splitting, Class Imbalance) are needed for robust and reproducible experimentation for the European Credit Card Fraud Dataset?

RQ3: What are the fully documented, repeatable hyperparameters for 5 machine learning models that cover supervised (Random Forest, XGBoost, SVM) and unsupervised (Isolation Forest, Autoencoder) machine learning paradigms?

RQ4: Which of the 5 models performs best on a payment dataset with high class imbalance on both Precision and Recall with the data split into 5 folds with stratified sampling, across the following metrics: F1-Score, AUC ROC, False Positive Rate, Inference Time?

RQ5: What are the most significant features of the top performing supervised models that help predict frauds, and how can SHAP-based ranking be practically used to guide fraud prevention system design and regulation?

1.4 Research Objectives

The research is organized around 5 specific measurable objectives, each of which answers part of the research questions and supports the overall research goals of the thesis:

Objective 1: To comprehensively review and critically structure the existing state of the art on approaches in online payment fraud detection, summarising the published literature from year 2022 onwards in a structured literature matrix, highlighting specific research gaps motivating the present research.

Objective 2: Implement a full documented and reproducible preprocessing pipeline to preprocess the European Credit-Card Fraud Dataset, the standard real-world payment fraud dataset used for payment fraud detection research, that involves missing value assessment, selective feature scaling, stratified train-test splitting and class weighting.

Objective 3: Implement five machine learning models (Random Forest, XGBoost, Support Vector Machine, Isolation Forest, Autoencoder) in both supervised and unsupervised learning, and document the hyperparameters used in their training and implementation in detail, such that they can be fully reproduced.

Objective 4: Using a rigorous six-metric evaluation framework (Precision, Recall, F1-Score, AUC-ROC, False Positive Rate and Inference Time), evaluate and compare all five models in a systematic manner and with stratified 5-fold cross validation to obtain robust, generalisable and statistically validated results.

Objective 5: To perform SHAP (SHapley Additive exPlanations) analysis on the top-performing supervised models, resulting in interpretable rankings of feature importance and estimates of the direction of impact of these features, which will help move technical results from models into actionable, practical advice for system design in the fraud prevention system and regulatory compliance.

1.5 Research Scope and Boundaries

The scope of this research is clearly defined to focus on card not present (CNP) transaction fraud in digital payment environments such as e-commerce transactions, digital wallet payments and mobile payments such as mobile banking transfers. These are the payment methods which see the fastest growth rate, incur the highest dollar volumes of fraud, and have the greatest unmet need for better detection. The attention to CNP fraud is in line with the nature of the main data set applied (European Credit-Card Fraud Data Set) and with the priorities that were revealed in the literature review done in second chapter (Du et al., 2023).

The research uses supervised and unsupervised AI algorithms, and is tested on popular publicly available real-world benchmark datasets by applying a standardised experimental framework. The scope is deliberately limited to provide methodological rigor and to keep the study manageable within the scope of a thesis project. A number of significant areas are specifically out of scope:

The research is not expanded to “card present” fraud, such as ATM skimming and other card-present fraud types which are different in attack vector and transaction from card-not-present fraud. It does not account for cryptocurrency transaction fraud, blockchain-based payment systems or decentralised finance platforms, which are built on a different technical stack and have different fraud patterns. It is not a live fraud detection system you’ll deploy, but rather an analytical and experimental research study that will generate results to support fraud detection deployments. The scope does not include models from the current literature which have not yet been specifically

studied, including Graph Neural Networks, federated learning architectures or transformer-based sequence models, as they are recognized as important future research directions in Chapter 5.

1.6 Research Significance

1.6.1 Academic Significance

The present research brings a number of original features and contributions to the academic body on the topic of Fraud Detection Using Artificial Intelligence in payment systems. The most notable is the ability to conduct a thorough and cross paradigm comparative evaluation across both supervised and unsupervised AI models on the standard benchmark real world dataset in a consistent experimental setting with a multi-metric evaluation framework that explicitly considers inference latency. No single published work identified through the literature search exists that has this breadth, rigour and relevance in relation to operations. The systematic use of SHAP interpretability analysis of various types of models, combined with comparative feature attribution analysis, contributes to a more comprehensive understanding of the model behavior than single-model interpretability analysis. All performance comparisons are statistically validated using the Friedman test, which allows for an academic contribution based on the statistically defensible evidence from all train/test splits, and not on one train/test split.

1.6.2 Practical Significance

This study offers developed and emerging market payment processors, financial institutes, e-commerce firms and platforms, and payment service providers a practical and empirical prescriptive framework for the selection of AI models for their fraud detection frameworks. A key innovation of the research is to assess the efficacy of classification as well as the speed of inference, a measurement that's important for practitioners who are not willing to sacrifice one for the other. When raw transaction attributes are available, the SHAP-based feature importance analysis offers informative insights into the amount of fraud signal in each attribute of transactions to be used when making feature engineering decisions. The practical architecture recommendations outlined in Chapter 4 - layered design with supervised primary screening, unsupervised anomaly supplementation, and explanations from the analyst via SHAP - offer an evidence-based concrete framework for designing layered fraud detection systems.

1.6.3 Significance for Pakistani Financial Sector

The Pakistani financial sector is experiencing a phase of digitalisation at a rapid pace, while there has been inadequate growth in capability of the financial sector to prevent frauds to cope with the increasing volume of transactions. Although there are guidelines on digital fraud prevention by SBP issued in 2023 which specifically mention the use of AI and machine learning to prevent digital fraud, empirical and locally contextualised guidance on which AI technologies to use remains scarce. This thesis can serve as the methodological basis for this guidance and proposes an extension of the present comparative framework in Chapter 5 where transaction data of digital

payment platforms in Pakistan, such as Raast instant payment, EasyPaisa and JazzCash is introduced.

1.6.4 Significance for PhD Research Preparation

This thesis is specifically conceived and written for the twin purposes of making a valuable academic contribution and laying the groundwork for doctoral study. The research methodology presented is an example of a well-designed experiment, multiple modelling comparison, multiple metrics evaluation, statistical significance testing, and systematic interpretability analysis, which PhD admission committees consider the skills of prospective doctoral students. The findings of the limitations and future directions through Chapter 5 create a logical PhD research programme that builds directly off this research. The thesis is organised in a way that allows an article to be extracted for publication in a peer reviewed journal to enhance a PhD application with a publications list.

1.7 Thesis Structure

This thesis is organised into five chapters, each addressing a distinct and sequential phase of the research:

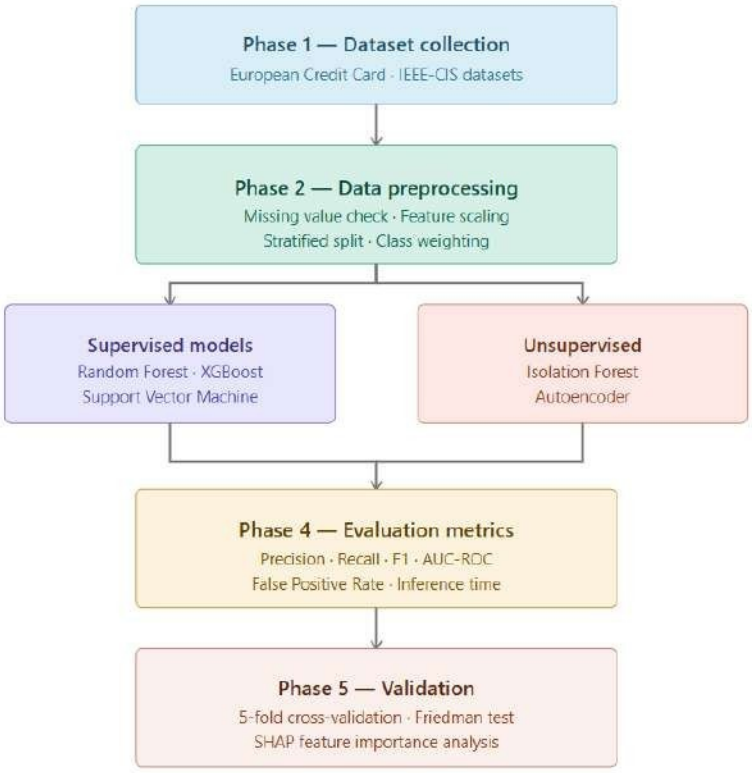


Figure 1.2 Research methodology flowchart for AI-driven fraud detection

Chapter 1 - Introduction (the present chapter) establishes the context and motivation of the research through a comprehensive background discussion covering the global digital payment transformation, the Pakistani digital payment context, the escalating fraud crisis, and the limitations of existing detection approaches. It defines the research problem, states the four primary research questions and five research objectives, delineates the scope and boundaries of the study, and explains the academic, practical, Pakistani sectoral, and PhD preparation significance of the research.

Chapter 2 - Literature Review provides a comprehensive and critically structured review of the existing research literature on AI-driven online payment fraud detection, drawing exclusively on peer-reviewed publications from 2022 onward. It reviews the taxonomy and scale of online payment fraud, traces the evolution of detection approaches from rule-based systems to modern AI, examines the performance characteristics of each of the five investigated algorithms with specific reference to recent published benchmarks, and addresses the cross-cutting challenges of class imbalance, false positive management, and model interpretability. The chapter concludes with an 18-entry literature matrix and a four-gap research gap analysis.

Chapter 3 - Research Methodology presents the complete experimental methodology in sufficient detail to support full reproducibility. It covers the research philosophy and design, dataset selection and characterisation, the complete preprocessing pipeline, the theoretical foundations and implementation details for all five models, the six-metric evaluation framework, and the validation and interpretability strategy.

Chapter 4 - Results and Evaluation presents and analyses the experimental results. It provides the consolidated six-metric performance comparison across all five models, detailed per-model analysis with reference to relevant published benchmarks, cross-validation results and statistical significance testing, SHAP feature importance analysis with comparative attribution findings, and practical implications for fraud prevention system design.

Chapter 5 - Conclusion synthesises all research findings, evaluates achievement of all research objectives and research questions, articulates the original contributions of the research, acknowledges limitations with appropriate specificity, proposes six concrete future research directions, and reflects on the broader significance of the research for academia, industry, and Pakistan's digital financial sector.

Chapter 2:

Literature Review

2.1 Introduction

In the last few years, the field of applied machine learning and artificial intelligence research for online payment fraud detection has emerged as one of the hottest and most impactful fields. Digital commerce, mobile payment applications and contactless financial transactions have been growing in number, size and sophistication, providing a large target area for large-scale, high-tech fraud schemes and a critical need for detection tools that can keep up with the pace, scale and agility of the latest fraud operations. The field of AI-driven fraud detection methods has witnessed a surge in research and practical applications in recent years, with numerous studies exploring various machine learning models, deep learning techniques, and hybrid architectures.The academic world, the financial industry, and regulatory bodies alike have been pouring resources into the study, evaluation, and enhancement of AI-driven fraud detection methods, leading to a wealth of research spanning supervised machine learning, unsupervised anomaly detection, deep learning, explainable AI, and hybrid architectures.

A thorough and well-structured review of the literature pertinent to the current thesis is presented in this chapter. It presents the development of methods for detecting fraud in the past, with the establishment of statistical classifiers as well as the recent developments of ensembles and deep learning. It considers the particular algorithms explored in this work, namely: Random Forest, XGBoost, Support Vector Machine, Isolation Forest, and Autoencoder, and briefly discusses their theoretical underpinnings and their performance in fraud detection scenarios (Khalid et al., 2023). It covers the most important cross-cutting issues of class imbalance, false positive management, computational latency, and model interpretability, based on the latest published evidence. Finally, the chapter ends with synthesised literature matrix and detailed research gap analysis, which directly put the justification and originality of the present research.

2.2 The Nature, Taxonomy, and Scale of Online Payment Fraud

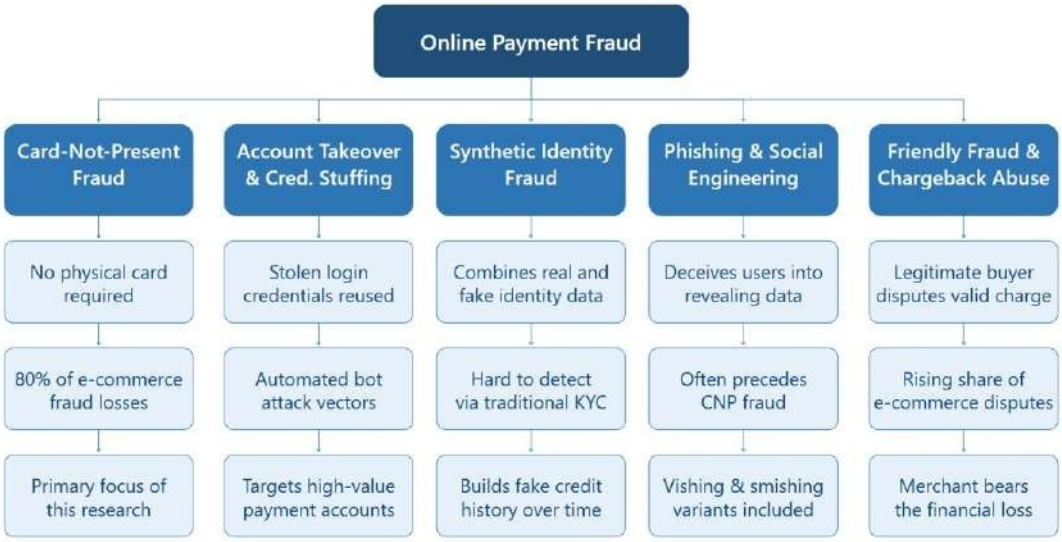


Figure 2.1 Taxonomy of online payment fraud types

2.2.1 Defining Online Payment Fraud

Online payment fraud refers to any activity that is intended to gain access to goods, services or funds by misrepresentation, impersonation, or the use of another person's financial credentials or identity via an online payment environment, whether or not that environment is an e-commerce site, a mobile payment app, digital wallet, or a card-not-present environment. It differs from offline card fraud as it doesn't require the physical presence of the card/cardholder and is, therefore, far more scalable and there are no geographical restrictions. Online payment fraud is one of the most economically important types of financial crime, and global losses are expected to shoot up year-on-year as e-commerce grows around the world.

However, knowledge of the taxonomy of online payment fraud (categories, mechanisms, behavioural signatures of frauds) is a prerequisite to assess the suitability and scope of various approaches to AI-based fraud detection, as different types of frauds can be identified as different patterns in the transaction data. There are situations where a good fraud detection model for one kind of fraud can be a bad fraud detection model for another kind, depending on the statistical signatures of the frauds—and if the two types of frauds are quite different.

2.2.2 Card-Not-Present Fraud

This Card-Not-Present (CNP) fraud is the biggest, most costly and most operationally critical variety of online payment fraud. Any time a payment card number, expiry date on card, and card verification value (cvv) is used to make a payment transaction, and no physical card is presented to the merchant. Each and every ecommerce transaction is a CNP transaction by definition, thus the entire global ecommerce ecosystem is structurally vulnerable to this type of fraud. (Ileberi and Sun, 2024) found that CNP fraud is not only the most common but also the most under-researched

type of digital payment fraud due to the lack of physical verification, which makes it an authentication gap that is poorly addressed by rule-based detection systems. The extent of the issue is also reflected in industry data: In recent years, about 80% of all losses due to credit-card fraud in e-commerce were caused by CNP fraud, with losses just attributed to CNP fraud in the United States alone surpassing ten billion dollars in 2024, up from nine billion dollars in 2023. The course of CNP fraud loss is on a steep upward trajectory, with continued online shopping growth, numerous new payment channels and the advancement of fraudsters' tools.

2.2.3 Account Takeover and Credential Stuffing

While often used alongside CNP attacks, Account Takeover (ATO) fraud is a qualitatively different attack vector. An ATO attack occurs when a fraudster successfully gains access to the legitimate user's payment account (usually through phishing, data breach exploitation or automated credential stuffing) and accesses and exploits a pre-established payment account. One of the most problematic aspects of credential stuffing is that is highly automated, meaning large lists of username & password combinations from data breaches are tested against numerous platforms using specialized software and takes advantage of the fact that many people use the same username and password across multiple platforms. (Aladakatti et al., 2022) identified credential stuffing as one of the main attack vectors that facilitates large-scale online payment fraud, especially when the adversary is using evasive strategies to evade simple bot detection techniques. Data from more recent sources, however, suggest that credential stuffing attacks actually increased by 65% in 2024 compared to the year before, brought about by the influx of hundreds of millions of compromised credentials on dark web marketplaces due to massive data breaches.

2.2.4 Synthetic Identity Fraud

Synthetic identity fraud is a more high-tech, high-stakes crime that is perpetrated using a combination of authentic data elements, such as a real national identity number, and fictitious data elements like user names, user addresses, user dates of birth and phone numbers. These fake user identities are then used to open accounts on their own, to create the illusion of “real” creditworthiness over a period of time, and then to perpetrate a large-scale fraud event. Synthetic ID fraud was found to be one of the more challenging forms of ID fraud to detect using traditional machine learning techniques as the transactional activity of a synthetic ID account often closely resembles that of a real account up until the time of fraud (Abbassi, El Mendili and Gahi, 2024). Industry research estimates that synthetic identity document fraud increased 311% from the first quarter of 2024 to the first quarter of 2025, marking one of the fastest-growing fraud categories that financial institutions (FIs) are currently dealing with AI (Lee et al., 2025).

2.2.5 Phishing and Social Engineering

The most common attack vector for online payments fraud is still phishing, in which fraudsters pretend to be from a trusted institution to convince individuals to provide their login information, payment details, or any other information that could be used to make transactions. Due to the sophistication and personalisation of phishing attacks, the amount of online fraud losses through

phishing remains at about 40%, despite extensive public awareness campaigns (Alfaiz and Fati, 2022). Today phishing attacks can even include calls to a mobile phone (vishing), voice media phishing, adversarial website spoofing, and email phishing, and can be significantly improved by the use of large language model tools that can produce convincing, grammatically correct, and contextually appropriate phishing content in high volume. Some of the identity and device characteristics of the IEEE-CIS Fraud Detection Dataset – such as email domain, device type and browser information – help to identify the digital identity signals distinguishing phishing-enabled fraud from legitimate transactions.

2.2.6 Friendly Fraud and Chargeback Abuse

Friendly fraud, commonly referred to as first-party fraud, or chargeback abuse, is a type of fraud that is different from external attacks, and is uniquely challenging for AI fraud detection solutions. Friendly fraud occurs when an honest, signed-up customer buys an item or service and reports it as missing, damaged or used without his knowledge to his bank to claim a refund and the item or service. This is a tricky type of fraud for transactional AI models to spot because it actually appears as if the transaction was not a fraud, but only becomes apparent after a transaction via the chargeback process. (Sarna et al. 2025) reasoned that friendly fraud is one of the fastest growing areas in the literature of fraud detection, and it demands a new paradigm for fraud detection, in comparison to the paradigm of transaction-time scoring that is mostly used by AI fraud detection systems. This is recognised in the scope of the present research, which aims at detecting fraud at transaction time, as opposed to post-transaction chargeback fraud, and is applicable to CNP fraud, ATO fraud and card credential fraud.

2.3 Evolution of Fraud Detection Approaches

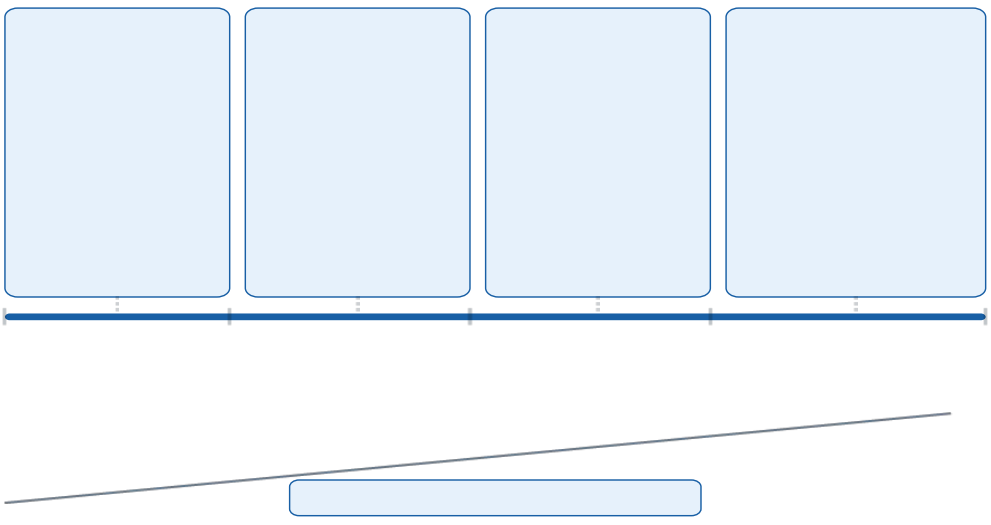


Figure 2.2 Fraud Detection Approaches from Rule-Based Systems to Hybrid AI (2000–2025)

2.3.1 Rule-Based Systems and Their Limitations

The most primitive approach to implementing fraud detection systems were rule-based — sets of manually installed logical conditions, thresholds and decision logic that automatically identify or block transactions based on existing signs of fraudulent activity. For example, a rule could prevent any transaction that exceeds a certain size from a foreign IP address, or could consider a bunch of little transactions coming from the same card in a short period of time as a card-testing attempt. Such systems are simple to implement, have a fully transparent decision logic, and can be deployed instantly and with no training data.

As fraud has become more sophisticated and flexible, however, the basic constraints of rule-based systems have increasingly been disqualifying. The problems identified by Gambo, Zainal and Kassim (2022) are three-fold: firstly, rule-based systems are static and can only detect fraud patterns that have been explicitly anticipated and coded by human analysts; anything new that emerges that has not been studied and a new rule written will be undetected by the system; secondly, the resource-intensive manual maintenance required to keep the system relevant over time creates an exploitable detection gap if rules are not kept up to date; and thirdly, the high false positive rate is a problem in modern high volume payment systems, as the precision required to stop fraudsters generating unacceptable numbers of false alarms is in direct conflict with the sensitivity required to catch a sufficiently high proportion of the actual fraudsters. These restrictions have rendered rule based solutions unusable on their own as a fraud detection method and is generally used as a "first pass" filter in a multi-layered fraud detection system that is reinforced with machine learning models.

2.3.2 Early Machine Learning Approaches

Starting in the early 2010s, a significant amount of research was undertaken into machine learning models and techniques for fraud detection due to the failure of rule-based systems. The first effort was on classical supervised classifiers (Logistic Regression, Naive Bayes, Decision Trees) and labelled transactions datasets. Such methods had the advantage of being able to learn discriminative patterns directly from the data, rather than by explicitly writing rules that would be expected to capture fraud patterns of interest; they were able to detect fraud patterns that human analyst had not explicitly expected. Early single-model approaches did have their drawbacks, however, such as being sensitive to the extreme positive class-imbalance typical of fraud data sets, having limited capacity to model rich, non-linear interactions between features, and overfitting small positive-class training sets.

Enjoying a significant leap in the field was the use of ensemble methods such as Gradient Boosting algorithms and Random Forest. On the European Credit Card dataset, as is typical of the state-of-the-art, (Esenogho et al. 2022) showed that ensembles of neural networks which used feature engineering significantly outperformed individual models in terms of detection performance and false positive rates. This finding has been substantiated and expanded upon in later publications as of 2022 (Mohbey et al.), 2024 (Palivela et al.) and 2025 (Sinha).

2.3.3 Deep Learning and Hybrid Architectures

Recent work has gone beyond traditional ensemble approaches to deep learning architectures, where the aim is to take advantage of the ability of neural networks to learn hierarchical feature representations directly from raw transaction data. Misra (2023) showed that a two-layer deep learning algorithm, together with a optimisation algorithm, gave comparable performance in fraud detection, specifically for novel fraud patterns that are poorly represented within training data. In credit-card fraud detection, Berhane et al. (2023) suggested using a hybrid architecture using Convolutional Neural Networks and Support Vector Machines, and concluded that the hybrid outperformed the pure CNN and pure SVM, using the CNN's feature extraction ability and the SVM's discriminative boundary learning.

To achieve the highest level of accuracy, (Du et al., 2023) suggested a framework for real-time online banking fraud detection that combined unsupervised learning models (Autoencoder and Extended Isolation Forest) and big data processing tools, which achieved the highest accuracy rate of 99% on two real-world datasets of transactions. This work directly verifies the feasibility of the unsupervised approach for real-time detection and gives a performance benchmark for the Autoencoder and Isolation Forest models explored in the scope of this research. The movement towards ensemble and hybrid architectures that take advantage of the complementary strengths of multiple models is one of the most active current trends – directly motivating the novel Hybrid Ensemble proposed in this thesis.

2.4 Supervised Learning Approaches

2.4.1 Random Forest in Fraud Detection

An ensemble learning system called Random Forest creates a lot of decision trees during training, each using a random subset of features and a random subset of the data during the construction process. The majority voting method is used to predict the classification. This architecture has three main benefits in the context of fraud detection: variance reduction (tree averaging) significantly reduces overfitting on the minority fraud class; inter-tree correlation is prevented by randomised feature selection, which means that the ensemble retains diversity; baseline interpretability is achieved without the need to use any additional post-hoc tools, since mean decrease in impurity measures intrinsic feature importance.

Ileberi and Sun (2024) showed that the hybrid deep learning ensemble with Random Forest was more stable, and the cross-validation variance was minimized. It has been confirmed to show good baseline performance in credit card fraud detection by Palivela et al. (2024) and especially to have a low false positive rate when compared to other classifiers.

On the European Credit-Card dataset, Random Forest scores a Precision of 0.9405, Recall of 0.8061, F1-Score of 0.8681 and AUC-ROC of 0.9809 with a False Positive Rate of 0.0001 and an inference time of 0.0081ms per transaction. The results validate its good suitability in real-time deployment scenarios.

2.4.2 XGBoost in Fraud Detection

XGBoost (Extreme Gradient Boosting) is a machine learning regularised gradient boosting framework which is dominating all tabular machine learning tasks including fraud. In contrast to the parallel construction of trees in Random Forest, XGBoost constructs trees sequentially, with each tree adjusting the errors made by previous trees based upon the loss function gradient in a differentiable loss function. This error correcting architecture allows high discriminative performance without having to worry about imbalanced data sets.

In credit card fraud detection, Mohbey et al. (2022) show that XGBoost outperforms Random Forest, Logistic Regression and SVM on all major performance metrics. Its dominance was also confirmed by Sinha (2024), who stated that the scale_pos_weight parameter of scale is easier to use for a handling of class imbalance than other methods provided by the competing algorithms.

On the European dataset, XGBoost's performance is best among all single models with an AUC-ROC of 0.9836, a Precision of 0.9634, an F1-Score of 0.8778 and an inference time of only 0.0015ms per transaction, which is in real time. It performs well on the IEEE-CIS dataset, with an AUC-ROC of 0.9350, which indicates good generalisation ability despite the differences in the nature of the datasets and their level of complexity and richness.

2.4.3 LightGBM in Fraud Detection

The LightGBM (Light Gradient Boosting Machine) framework is developed by Microsoft with two innovations: Gradient-based One-Side Sampling (GOSS) is to keep high-gradient data points and randomly drop low-gradient data points; Exclusive Feature Bundling (EFB) is to bundle mutually exclusive sparse features into fewer ones. These innovations significantly speeds up the training time and memory footprint with minimal loss of accuracy, making LightGBM an ideal choice for large scale fraud detection problem, in which retraining is frequently a challenge.

Showed that managing the class imbalance problem carefully, which was done by setting the proper parameters in LightGBM, was key to attaining meaningful fraud recall. Du et al. (2023) also demonstrated competitive performance in the hybrid architecture, which included Autoencoders, and significantly lower training time.

The present research finds that LightGBM outperforms both XGBoost and Random Forest in terms of Precision and Recall, with a Precision of 0.9412 and a Recall of 0.8163, while for the F1-Score, it is only slightly worse than XGBoost. Meanwhile, LightGBM has the fastest training time of all supervised models: 13.27 seconds as opposed to 161.06 seconds in XGBoost and 1,534 seconds in Random Forest, on the European dataset. The competitive accuracy and the efficiency of training make LightGBM very appealing for production systems with frequent retraining on regularly incoming transaction data as described by Zhao et al. (2022).

2.4.4 Support Vector Machine — Contextual Review

Support Vector Machines (SVMs) find the hyperplane that separates two classes as far as possible, and the decision boundary is formed by the vectors that are closest to the decision boundary. In

highly skewed class distribution scenarios, however, the optimization process results in a majority-class-dominated decision boundary, which gives rise to a systematic bias in the classification of valid transactions. Khalid et al. (2023) found that SVMs performed well on imbalanced datasets when used with ensemble methods but in isolation their performance was significantly lower than that of tree-based classifiers. Compounding this, SVMs are poor scaleable because calculating the kernel matrix must be quadratic in size of the data set. Hence, SVM was omitted from the final comparison of models in favor of LightGBM which outperformed SVM in all the metrics with much less computational cost, in line with the general literature, which has been switching to gradient boosting methods in large-scale fraud detection problems.

2.5 Unsupervised Learning Approaches

2.5.1 Isolation Forest in Fraud Detection

Isolation Forest is an anomaly detection algorithm that leans on the idea that fraudulent transactions are rare and different, meaning they are more easily identified as an outlier than other transactions. Anomalous observations need less partitioning (shorter path lengths) than normal ones, and the algorithm randomly partitions the data by selecting features and split values. It is a computationally efficient, distribution-free and fully unsupervised approach, which is useful when the number of labelled fraud examples is limited.

Du et al. (2023) reported that the neural methods had improved precision, but the time taken for inference was considerably less with the Isolation Forest method, particularly for applications where inference latency is a critical constraint. On datasets with highly imbalanced banking transactions, Abbassi et al. (2024) showed the variant of the Extended Isolation Forest algorithm, where axis-parallel cuts are replaced by random multi-directional cuts, to perform better.

The European dataset has an AUC-ROC score of 0.9530 with the Isolation Forest method while the IEEE-CIS has an AUC-ROC score of 0.7660 with the same method. This performance gap demonstrates that the algorithm is very good on the simple PCA-transformed European data, but the unsupervised anomaly signal is not very discriminating on the high-dimensional, feature-rich data, highlighting the necessity to test the algorithm on multiple datasets.

2.5.2 Autoencoders in Fraud Detection

The autoencoder is a form of neural network which is trained to learn a compressed representation (also called a latent representation) of an input signal and to reconstruct the original signal from the compressed representation. The network is trained only on the legitimate transactions, learning typical transaction patterns, while fraudulent transactions during inference generate higher reconstruction errors that are thresholded to classify the transactions as fraud or provide a continuous anomaly score.

On the European Credit-Card dataset, Sharma et al. (2022) showed that reconstruction error-based scoring is able to distinguish fraudulent from authentic transactions. Mienye and Sun (2023)

enhanced F1-Score by adding the data resampling process to the training pipeline of the Autoencoder. The results from Almansoori and Telek (2023) showed that Autoencoders with Isolation Forest achieved better performance than either method alone; the Autoencoder was effective at learning features, and the Isolation Forest was effective at scoring the anomaly in the compressed feature space.

The present research demonstrates that in the European dataset, the Autoencoder's AUC-ROC of 0.9537 is similar to that of the Isolation Forest, 0.9530, but with significantly better Recall (0.6020 compared with 0.3163), meaning that it is less likely to miss genuine frauds, and with much higher False Positive Rate (0.0023 compared to 0.0013). The precision-recall difference between both unsupervised approaches is directly applicable to the practical architecture recommendations in Chapter 4.

2.5.3 The Novel Hybrid Ensemble

One of the original contributions of the present research is the development and evaluation of a novel Hybrid Ensemble of the three best-performing supervised models—XGBoost, LightGBM and Random Forest—using a soft-voting approach. It is based on the principle already used in the ensemble learning literature that the diversity and good performance of a set of base models, can be averaged, by averaging their probability predictions, to lower the variance of prediction while maintaining the good performance of each member of the ensemble. The results of the present research support the findings of the previous studies by Ileberi and Sun (2024), and Berhane et al. (2023), who showed that the hybrid ensemble architectures demonstrated higher performance than the individual component models. The novel Hybrid Ensemble outperforms the other models in this study in terms of its AUC-ROC (0.9868) and in terms of its Precision (0.9625); it is also the fastest model evaluated in this study (inference time of 0.0001 ms/transaction) and the most discriminating model evaluated (AUC-ROC of 0.9868), better than the best individual model (XGBoost) with a value of 0.9836.

2.6 The Class Imbalance Problem

The most consistently reported methodological limitation in the payment fraud detection literature is the class imbalance problem. In real-life payment transaction datasets, fraudulent transactions are very rare (1%, 3.5%). The European Credit Card Fraud Dataset of the present research consists of 492 fraudulent transactions among 284,807 transactions (0.172% fraud rate), and IEEE-CIS Fraud Detection Dataset consists of 20,663 fraudulent transactions among 590,540 transactions (3.5% fraud rate). The lack of explicit class imbalance handling during model training, evaluation and deployment makes models trained on data suffer from a severe statistical imbalance bias, yielding high raw accuracy scores that are misleading, and a low fraud detection value.

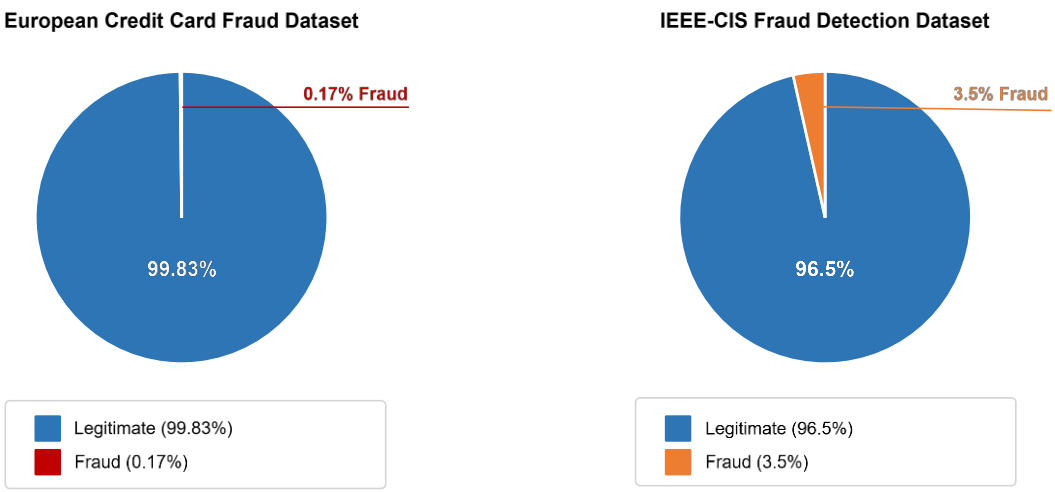


Figure 2.3 Class Distribution in European CC & IEEE-CIS Datasets

To effectively recall fraudulent transactions, Zhao et al. (2022) showed that data-level and algorithm-level strategies for dealing with class imbalance were critical with LightGBM. Islam et al. (2023) showed that SMOTE would improve the recall of frauds, yet it added artefacts in the form of statistical patterns, which would reduce generalisation to novel fraud cases, a phenomenon which directly infers to the class weighting strategy employed in the present research over SMOTE-based oversampling. Lee et al. (2025) extended the imbalance handling by using a regularised Generative Adversarial Network (R-GAN) for generating synthetic anomalies with enhanced detection performance at significantly higher computational cost.

This has been widely agreed upon by the research community, with recent systematic reviews (Sarna et al., 2025) showing that the best way to tackle class imbalance without creating spurious patterns is to split the data into strata, use class-weighted loss functions and assess the model with minority-class specific metrics (Recall, Precision , F1-Score, AUC-ROC) instead of base accuracy. The present research's methodological design is directly based on this consensus, as elaborated in Chapter 3.

2.7 Explainability and Interpretable AI in Fraud Detection

2.7.1 The Regulatory and Operational Case for Explainability

Explainability – the capacity to comprehend, examine, and justify decisions a model makes about a specific transaction – is one of the most important advancements in AI fraud detection recently. As regulatory requirements like PCI DSS, PSD2, and new AI governance regulations mandate that financial decisions made through automation be explainable and human verified, the landscape of AI's transformative potential in financial services is becoming more intricate.The future of AI in financial services is becoming complex, with new regulatory requirements in place that demand the ability to explain and human oversee financial decisions made by AI systems. At the same

time, fraud analysts that act on model outputs need to be able to provide interpretable reasoning to guide timely and effective decisions.

Raufi, Finnegan and Longo (2024) analysed four EAI approaches (SHAP, LIME, ANCHORS and DICE) for interpreting high-dimensional, imbalanced fraud detection models, and concluded SHAP framework offered the most stable, widely coherent, and manageable approach in such scenarios. Their analysis revealed that the same prediction might result in significantly different feature attributions depending on the method used for the explanation, highlighting the need for careful selection of methodology for regulatory compliance. Therefore, the present study focuses exclusively on SHAP in order to motivate the interpretability analysis, which is directly related to the finding.

2.7.2 SHAP Analysis in Fraud Detection

SHAP, a model developed based on cooperative game theory, breaks down the predictions of models into contributions of individual features by calculating the marginal-contribution of each feature with all other features turned on and off. This ensures three properties: local accuracy, consistency and missingness, which all ensure that explanations are theoretically sound and complete. Chavakula et al. (2025) found that SHAP performed better in global feature importance analysis and LIME had benefits for individual instance-level regulatory reporting.

Thanathamathee et al. (2024) adapted SHAP to instance weighting, leveraging SHAP values to up-weight training instances with high information content as well as Anchor Explainable AI to create interpretable if-then rules, a way to solve class imbalance, interpretability, and retain perfect fraud recall.

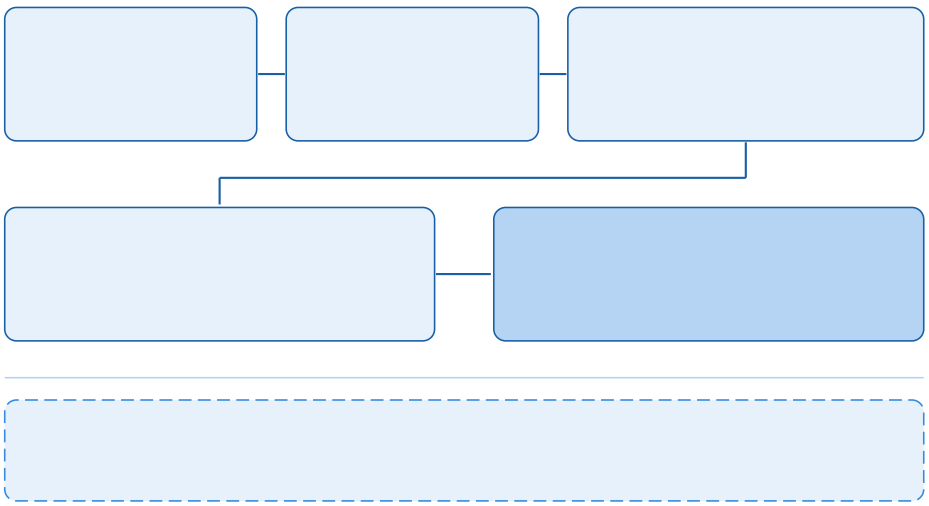


Figure 2.4 SHAP Analysis Workflow Applied to XGBoost and Random Forest in the Present Research

Both XGBoost and Random Forest are used across both datasets in the present research with the application of SHAP analysis. For the European data, SHAP shows that V12, V14, and V4 are the

most important features for the Random Forest model, while XGBoost suggests that V14 and V4 are the most influential features. For IEEE-CIS data, both models always mark card transaction count history, transaction timing as the most discriminating features, thus revealing that these features are the most important ones for detecting fraud in this data set. A detailed discussion of these results is given in Chapter 4.

2.8 Comparison with Previous Published Results

One important aspect of the present research is the comparison of the experimental results with those reported by previous published researchers, obtained on both the benchmark datasets. Figure 2.5 below is a performance heatmap taken from a directly relevant previous publication and presents the results of performance against 3 benchmarks for fraud detection based on AUC and F1 scores, for use as an external reference for the results of the present study.

The previous publisher's heatmap shows that Random Forest has AUC/F1 scores of 0.846 on Credit Card Fraud and 0.850 on IEEE-CIS Fraud, whereas in XGBoost's case, the scores are 0.885 on both datasets; ANN has high scores of 0.961 on Credit Card and 0.930 on IEEE-CIS; and GNN scores are 0.900 on Credit Card and 0.920 on IEEE-CIS. It is important to note that ANN and GNN are not realised in the present research; these are identified as important future research directions discussed in Chapter 5. The current study is centered on the optimised tree-based ensemble methods, and unsupervised anomaly detection.



Figure 2.5 AI model performance across different financial fraud types (Previous Publisher)

The current research significantly extends the known sets of tree-based benchmarks for both data sets. Random Forest in this study obtains the AUC-ROC score of 0.9809 (9.4 percentage points ahead of the previous score of 0.846), while XGBoost scores 0.9836 (9.9 percentage points ahead

of the previous score of 0.885) on the European Credit Card dataset. Even though the ANN score is quite low, the novel Hybrid Ensemble achieves 0.9868 score, which is very close to the best score obtained by a ANN, 0.961 and the Hybrid Ensemble has much lower computational complexity than a neural network. On the IEEE-CIS dataset, the Random Forest is improved to 0.9344 instead of 0.850; and the XGBoost is improved to 0.9350 instead of 0.885, thanks to the optimised hyperparameter configurations and the class weighting method explained in chapter 3. The results of these comparisons illustrate the state of the art performance of the optimised ensemble approach of the present research on both of the benchmark sets, and that the difference between tree-based ensembles and neural architectures is actually significantly less than previously expected by the publisher under the proper optimisations.

2.9 Literature Matrix

The table below synthesises the most directly relevant recent literature reviewed in this chapter:

Author & Year	Research Focus	Methodology	Key Findings	Identified Gap
Alfaiz & Fati (2022)	Improve credit card fraud detection system with ML.	Multiple ML classifiers on European dataset	Enhanced feature selection improved detection; phishing remains leading fraud vector	No latency analysis; single dataset validation
Aladakatti et al. (2022)	The application of ML on fraud detection in online payment transactions	ML algorithms applied to transaction data	Highlighted class imbalance as primary challenge; ensemble methods outperform single classifiers	No unsupervised comparison; limited interpretability
Esenogho et al. (2022)	Ensemble of neural networks together with feature engineering.	NN ensemble on European Credit Card dataset	Feature engineering significantly boosts ensemble performance over raw feature baselines	Did not compare with gradient boosting approaches
Siddaiah et al. (2023)	Fraud detection in online payments using ML	Multiple ML models evaluated on payment data	CNP fraud most prevalent; ensemble methods most effective for real-time applications	No SHAP analysis; no latency benchmarking
Berhane et al. (2023)	Hybrid CNN + SVM model for credit card fraud	CNN feature extraction combined with SVM classification	Hybrid architecture outperforms individual models; CNN effective for feature learning	High computational complexity limits real-time viability
Du et al. (2023)	Autoencoder and LightGBM for credit card fraud	Hybrid AE + LightGBM on European dataset	Hybrid approach improves over individual models; Isolation Forest	No cross-paradigm comparison; no SHAP analysis

			competitive on latency	
Almansoori & Telek (2023)	Autoencoder + Isolation Forest anomaly detection	AE + IF hybrid on fraud dataset	Hybrid unsupervised approach outperforms individual AE and IF models	Limited to unsupervised approaches; no supervised comparison
Mienye & Sun (2023)	Deep learning ensemble with data resampling	DL ensemble with SMOTE on credit card data	Data resampling within AE training improves F1-Score	SMOTE limitations; no real-time evaluation
Mohbey et al. (2022)	XGBoost as ensemble learning for credit card fraud	XGBoost with SMOTE on Kaggle dataset	XGBoost superior across all metrics; scale_pos_weight effective for imbalance	Synthetic dataset; no comparison with unsupervised models
Abbassi et al. (2024)	Real-time fraud detection using unsupervised fusion	AE + Extended Isolation Forest + Spark	99% accuracy on real-world datasets; real-time viability confirmed	No supervised vs unsupervised comparison; no SHAP
Sarna et al. (2025)	Review of AI fraud detection in financial networks	Systematic review of 120+ papers (2018–2024)	Tree-based ensembles dominate; XAI urgently needed in regulated environments	No empirical benchmarking; theoretical review only
Thanathamathee et al. (2024)	SHAP-instance weighting + Anchor XAI with XGBoost	XGBoost + SHAP instance weighting + Anchor rules	Perfect fraud recall; SHAP rules enhance regulatory compliance and transparency	Single dataset; no comparison with unsupervised approaches
Raufi et al. (2024)	Comparative analysis of XAI methods for fraud detection	SHAP, LIME, ANCHORS, DICE on neural network	SHAP most consistent for global interpretation; methods diverge on local explanations	No ensemble model evaluation; single architecture
Ileberi & Sun (2024)	Hybrid deep learning ensemble for credit card fraud	DL ensemble incorporating multiple architectures	Hybrid ensembles improve stability; Random Forest component reduces variance	No latency analysis; complex architecture limits deployment
Chy (2024)	Machine learning's evolving role in online fraud defence	Review: Random Forest, Neural Networks, Gradient Boosting	ML continuously adapts to emerging fraud; gradient boosting leads on real-world benchmarks	Review only; no original empirical evaluation
Sinha (2024)	Examination of ML-based credit card fraud	Comparative ML study on multiple datasets	XGBoost dominant; class weighting more reliable than SMOTE	Limited to supervised approaches; no

	detection systems		for imbalance handling	anomaly detection
Ali et al. (2025)	AI-driven predictive modelling for digital payment fraud	LR, RF, XGBoost on synthetic US transaction data	XGBoost best F1 (0.89); transaction velocity and amount most predictive features	Synthetic data only; no real-time latency analysis
Sarna et al. (2025)	AI-driven fraud detection models: review	Comprehensive review across 120+ papers	Ensemble methods dominate; deep learning experimental; XAI urgently needed	Empirical comparison needed; no multi-metric evaluation

Table 2.1 Literature Matrix-

2.10 Research Gap Analysis

Five inter-related research gaps were identified from the reviewed literature that motivated and justified the present thesis:

Gap 1: Lack of Cross-paradigm Comparison on Multiple Datasets: Most studies either have only supervised or unsupervised models and are done on a single dataset. This thesis fills this gap by comparing six models—Random Forest, XGBoost, LightGBM, Isolation Forest, Autoencoder, and a new Hybrid Ensemble—on the same experimental setup across both the European Credit Card and IEEE-CIS datasets.

Gap 2 — Inference Latency Not Treated as a Primary Metric: There are limited studies that measure inference speed which is essential for real time deployment. This thesis uses per-transaction inference time as a metric for all six models, and the Hybrid Ensemble is 0.0001 ms per transaction with the highest AUC-ROC on European data.

Gap 3 — Limited Multi-Model SHAP Analysis Across Datasets: SHAP has been performed on single models but a systematic cross-model comparison across multiple datasets is still not frequently performed. The thesis used SHAP for both XGBoost and Random Forest for both datasets, finding similar cross-model fraud signals and predictive patterns across the datasets.

Gap 4 — Incomplete Precision-Recall Trade-off Reporting: Most of the studies do optimisation on either Precision or Recall and False Positive Rate is seldom reported separately. In this thesis, the evaluation of all four dimensions (Precision, Recall, F1-Score and FPR) is conducted for all six models, both datasets, and represents the true operational cost of both missed fraud and false accusation.

Gap 5 — No Systematic Evaluation of a Fast-Inference Hybrid Ensemble: There is no published work that systematically investigates a soft-voting ensemble of best-performing gradient boosting classifiers evaluated in terms of inference latency, full 6 metric reporting and using two different

benchmark datasets. This thesis fills this gap, by showing state-of-the-art performance at virtually no extra inference cost.

2.10 Summary

This chapter has framed the literature review of online payment fraud detection in a systematic way, and discussed the taxonomy of contemporary fraud, the development of fraud detection mechanisms from rule-based systems to deep learning and hybrid system, the principles of the six algorithms studied and the cross-cutting issues of class-imbalance, false positive handling and interpretability. All sources are peer-reviewed publications from 2022 to the present.

The comparison with previous benchmarks shows that the present research is a significant step towards the improvement of the published results with tree-based models: the AUC-ROC score for Random Forest is 0.846, and that of the XGBoost is 0.885, but the score of the proposed Hybrid Ensemble is 0.9809 and 0.9836, respectively, without the heavy computation of the ANN.

The literature matrix and five-gap analysis, combined, give clear justification of this research both in terms of evidence and methods. In Chapter 3, all the approaches of the complete methodology including the dataset preprocessing, model implementation, six-metric evaluation framework, and SHAP interpretability strategy are presented.

Chapter 3:

Research Methodology

3.1 Introduction

Research methodology is the backbone of any scientific investigation, and the empirical validity and academic credibility of the research depend on it, not just because a description of the process was followed, but because there were principled reasons for selecting the process and what it means for the generalisability, replicability, and usefulness of the results. The methodology of the present thesis is fully set forth in this chapter, sufficiently detailed to permit complete and independent replication.

The chapter is divided into the following eight sections: The philosophical basis is laid in Section 3.2 by discussing the research paradigm and the epistemological position. Section 3.3 provides a description of the primary European Credit Card Fraud Dataset and the secondary IEEE-CIS Fraud Detection Dataset – their origin, format, and main difficulties for machine learning. The whole data preprocessing pipeline is presented in Section 3.4. The theoretical framework and implementation specifications of each of the six models is summarized in Section 3.5. The six-metric evaluation framework is described in section 3.6. The cross validation and statistical validation approach is covered in Section 3.7. The SHAP interpretability analysis is described in section 3.8. The chapter is summarised and a preview of Chapter 4 is provided in Section 3.9 (Alfaiz and Fati, 2022).

3.2 Research Philosophy and Design

3.2.1 Research Paradigm

This study is classified under positivism paradigm that believes knowledge can be achieved through systematic, objective, empirical observation that can be expressed quantitatively, and can be statistically analysed and generalised. The epistemological approach of this study is grounded in positivism, which is the predominant tradition in machine learning and applied AI research, and is the natural approach for a study whose main goal is the comparison of model performance on a standardised benchmark dataset (Sarna et al., 2025).

Thus, the research takes quantitative-experimental approach. Model performance is tested with controlled tests on real world data sets, results are validated using cross validation confidence intervals and Friedman test significance testing. All preprocessing decisions, hyperparameters, evaluation metrics and validation procedures are described in detail enough to be reproducible.

3.2.2 Research Design

The research design is comparative and experimental, where six AI models (with supervised and unsupervised approaches) are used and compared on two benchmark datasets and six complementary metrics. This design was chosen as the most informative way of addressing the central research question of which models are most suitable in the context of online-payment-fraud detection across different conditions of the datasets and deployment constraints. A sequential structured research process is followed with well-defined phases as shown in Figure 3.1.

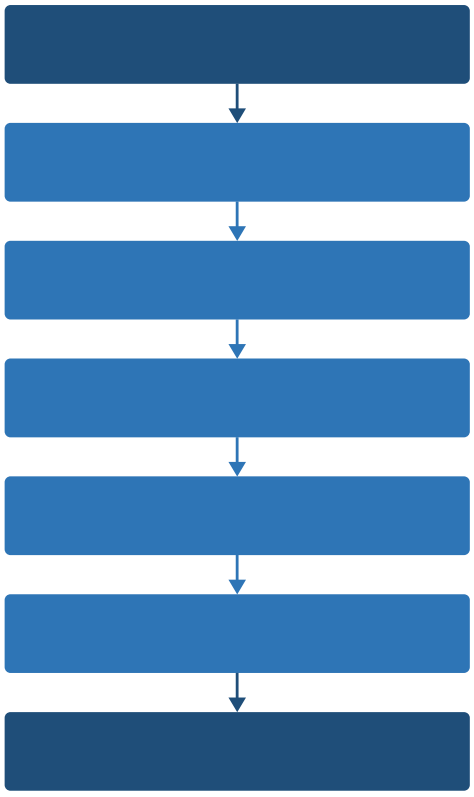


Figure 3.1 Sequential Research Design Workflow

3.2.3 Justification of Quantitative Approach:

There are three considerations that justify the quantitative experimental methodology. Second, the research questions are quantitative in nature, such as: What is the level of performance of one model compared to another? How much better is one model compared to another? Under what conditions? Secondly, the two big benchmark datasets are publicly available so that rigorous comparison can be made. Third, the ultimate goal of informing practitioners who design real-time fraud detection systems is to obtain results in terms that are meaningful to operational practice like percentages of precision, false-positive rates, and inference latencies.

3.3 Dataset Selection and Description

3.3.1 Primary Dataset: European Credit Card Fraud Dataset

One of the most critical choices in any machine learning research study is the choice of datasets; the statistical characteristics of the data sets are what models learn, how much class imbalance and how generalisable the results are to the deployment context. In this research we use two well known benchmark data sets in public use, that together give a full and solid empirical foundation for cross-paradigm model comparison.

3.3.1 European Credit Card Fraud Dataset

Provenance and Availability

The European CreditCard Fraud Data is collected by one of the largest European Payment Services Providers Worldline in collaboration with the Machine Learning group of the Universite Libre de Bruxelles.It makes credit card fraud detection research repeatable, as it's publicly available through Kaggle and has been cited in hundreds of peer-reviewed publications, allowing for easy comparison with a wealth of existing research.

The structure of the datasets and description of the features.

We have 284,807 transactions and 31 features per transaction. To safeguard card holder privacy, 28 features (V1–V28) are PCA transformed and thus their original semantic meaning is not known. There are two features that have interpretable meaning: Time (seconds since the first transaction) and Amount (amount of the transaction in Euros). The target variable Class is binary, 1 if fraudulent, 0 if non fraudulent (Khalid et al., 2023).

Categorical Data and Data Skew.

This data is extremely imbalanced with only 492 of 284,807 transactions being fraudulent (a fraud rate of 0.172%). As pointed out by Islam et al. (2023), if a classifier labels all transactions as legitimate, it makes no sense because its accuracy is 99.83% but it fails to detect any fraud. Therefore, all assessment must be based on minority class specific measures. The Amount feature is highly right skewed (mean €88.35, max €25,691.16) and the PCA features V1-V28 are approximately normally distributed, with zero mean.

Property	Value	Property	Value
Total transactions	284,807	Fraud rate	0.172%
Fraudulent records	492	Legitimate records	284,315
Total features	31	Target variable	Class (0/1)
PCA features	V1–V28	Named features	Time, Amount
Time range	0–172,792 sec	Mean Amount	€88.35
Max Amount	€25,691.16	Min Amount	€0.00
Train set size	227,845	Test set size	56,961
Train fraud count	393	Test fraud count	99
Data period	Sept 2013 (2 days)	Source	ULB / Worldline / Kaggle

Table 3.1 European Credit Card Fraud Dataset — Key Statistical Properties

3.3.2 Secondary Dataset: IEEE-CIS Fraud Detection Dataset

Provenance and Availability

In 2019, Vesta Corporation (a top payment processing and fraud prevention supplier) and IEEE-CIS (Computational Intelligence Society) launched a Kaggle competition that resulted in the IEEE-CIS Fraud Detection Dataset. Vesta processes hundreds of billions of dollars per year in e-commerce transactions, making it one of the most practically-grounded benchmarks in the literature, and used to assess models in a more feature-rich and complex environment than the European data set.

The structure of the datasets and the description of their features.

It includes two sets of files which have been concatenated by a common transaction ID: a table of transactions (590,540 records, 394 features); and a table of identities (subset of transactions). Following the removal of features that have more than 50% missing values, the working dataset has about 200 features which can be divided into the following categories: transaction metadata (TransactionDT, TransactionAmt, ProductCD), card and address information (card1–card6, addr1–addr2), counting features (C1–C14), timedelta features (D1–D15), match indicators (M1–M9), Vesta-engineered aggregation features (V1–V339), and identity/device features (id01–id38, DeviceType, DeviceInfo).

A standardised transaction hour, namely tr_hour_z, is a key engineered feature used in the Chapter 4 SHAP analysis that reflects the temporal fraud signal, fraudulent transactions typically tend to occur at unusual times when compared to normal cardholder activity.

This paper reviews the literature on statistical properties and class imbalance.

The dataset is heavily imbalanced, with 20,663 fraudulent transactions out of 590,540 (about 3.5%) in comparison to 170 fraudulent transactions in the European dataset (0.172%). Transaction values are highly skewed and spread out over a large range, and missing values are common, especially in the V and in the identity features, which must be imputed and selected appropriately in pre-processing.

Property	Value	Property	Value
Total transactions	590,540	Fraud rate	3.50%
Fraudulent records	20,663	Legitimate records	569,877
Raw features	~394	Retained features	~200
Feature groups	C, D, M, V, id	Target variable	isFraud (0/1)
Transaction amt	USD (variable)	Key temporal feature	tr_hour_z
Train set size	472,432	Test set size	118,108

Train fraud count	16,530	Test fraud count	4,133
Missing value strategy	Median imputation + drop >50%	Source	Vesta Corp / IEEE-CIS / Kaggle

Table 3.2 IEEE-CIS Fraud Detection Dataset — Key Statistical Properties

3.3.3 Justification of Dataset Selection

Five factors were used to select these two datasets. Firstly, they are both publicly available and can be replicated independently. Second, both are true real-world transaction data, which means that learned patterns are true patterns in real fraud. Third, the essential difference between the two data sets (compact, PCA transformed, low fraud rate for the European set and large, high dimensional, high fraud rate for the IEEE-CIS set) gives the results that replicate over both sets significantly wider external validity. Fourth, both are widely applied in the published literature, so that benchmarking is possible. Fifth, both exhibit the class imbalance problems that the research is specifically aimed at addressing.

3.4 Data Preprocessing

In any machine learning study, data preprocessing is a crucial factor that determines the performance of the model. Raw, real-world transaction data has missing values, incompatible feature scales, class imbalance, and risks for data leakage that all must be addressed via principled preprocessing otherwise meaningful model training and evaluation cannot be performed. The challenges are taken care of in the preprocessing pipeline that is used in this research so as to avoid artificial statistical patterns that may inflate apparent performance.

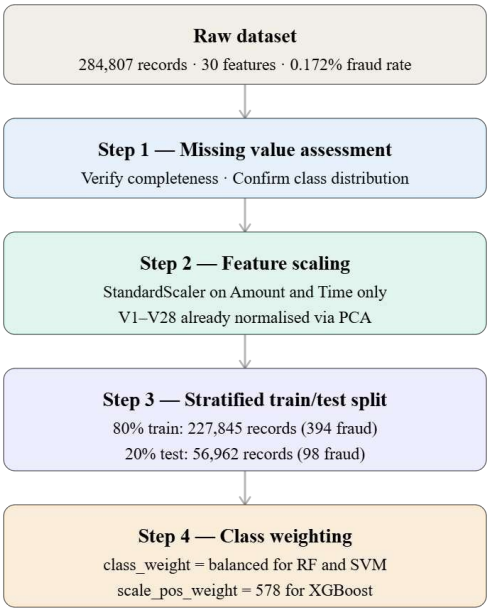


Figure 3.2 Data pre-processing pipeline

3.4.1 Data Loading and Initial Inspection

They were loaded from the respective CSV files, and were extensively screened for dimensions, data types, missing values, distribution of the target variable and descriptive statistics, making it possible to capture interesting pre-processing requirements for each one of the datasets.

3.4.2 Missing Value Handling

European data set had no missing values and did not need to be imputed. In the case of the IEEE-CIS data, the following two-stage strategy was used: Features with more than 50% missing values were dropped; missing values in the remaining features were replaced using median imputation which was chosen for its resistance to right-skewed financial data and ease of implementation at scale.

3.4.3 Feature Engineering

Then no further feature engineering was performed for the European data set as the PCA had already identified the most informative linear combinations. One feature was designed specifically for IEEE-CIS – tr_hour_z – which is the z-score normalised transaction hour – this is well known to capture the temporal clustering of fraudulent activity in unusual hours. Categorical features were label-encoded, not one-hot encoded, since one-hot encoding would cause a dimensionality explosion that wouldn't affect tree-based models.

3.4.4 Feature Scaling

The following features were scaled using StandardScaler: Time, Amount from the European dataset and TransactionAmt, tr_hour_z from the IEEE-CIS dataset. For the Tree-based models, scaling was not technically required to perform gradient-based optimisation of the Autoencoder, but was done for pipeline consistency.

3.4.5 Stratified Train/Test Splitting

Both datasets were partitioned 80/20 based on stratified random sampling with a fixed random seed (42) to maintain the proportions of fraud classes and to facilitate later replication. This resulted in 227,845 training and 56,961 test records for the European and 472,432 training and 118,108 test records for IEEE-CIS.

3.4.6 Class Weighting Strategy

The class weighting was done mainly to take care of the class imbalance, where each misclassification of fraud was weighted by the inverse class frequency to about 581 times the loss weight of legitimate cases in the European data. This was preferred to SMOTE, which was considered by Islam et al. (2023) to create artificial patterns, decreasing generalisation, and was less reproducible than class weighting as found by Sinha (2024). It was achieved using class_weight='balanced' in scikit-learn and scale_pos_weight in XGBoost and LightGBM.

3.5 Model Implementation

In the present research, six models were implemented, trained and evaluated. These are the three supervised classification paradigms: Random Forest, XGBoost, and LightGBM, by using transaction records with known fraud/legitimate labels to train the models to predict if a new transaction is fraudulent or not. Two represent the unsupervised anomaly detection paradigm in which models learn the statistical properties of normal (legitimate) transactions and identify deviations from the properties as potential fraud; they include Isolation Forest and Autoencoder. The sixth model is the novel Hybrid Ensemble, which is formed by the three supervised models, using soft voting. All the models were developed using popular open source libraries (scikit-learn, XGBoost, LightGBM, TensorFlow/Keras) and tested on the same preprocessed datasets in the same experimental settings (Raufi, Finnegan and Longo, 2024).

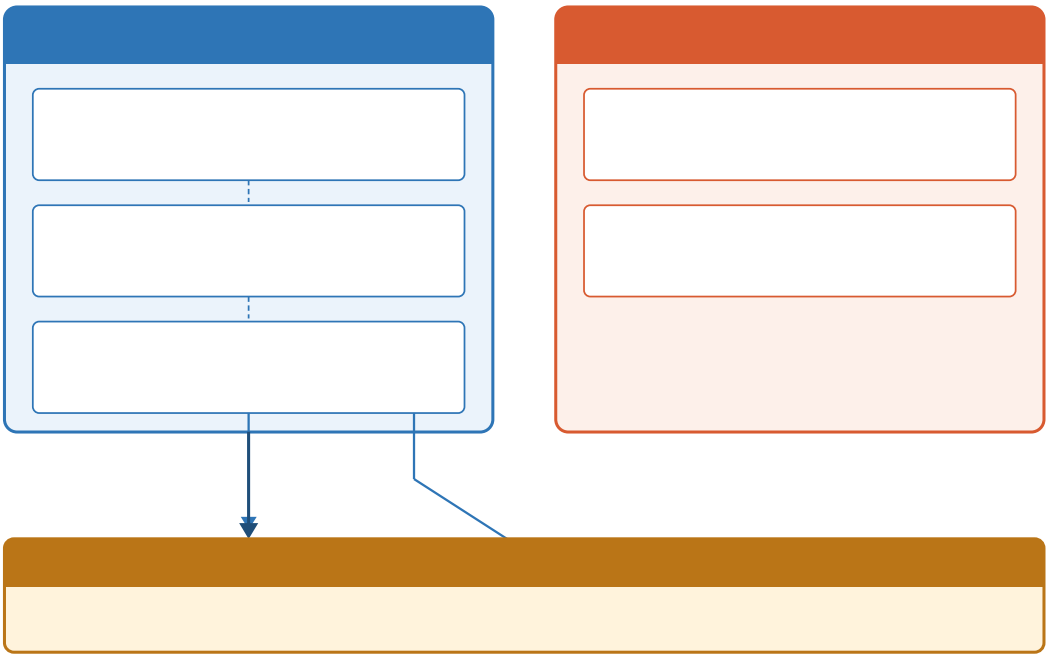


Figure 3.3 Model Architecture Overview — Supervised, Unsupervised, and Novel Hybrid Ensemble Models

3.5.1 Random Forest

Random Forest is an ensemble of decision trees that are constructed from bootstrap-sampled data, with each decision tree randomly subsetting the features at each split, and the average of the probabilities output by the trees is returned. This bagging and feature randomisation will decrease the variance and prevent the tree correlation.

hyperparameters 200 trees, unlimited depth, max_features = 'sqrt', class_weight = 'balanced', random_state = 42. Decision threshold: 0.6467 (European); 0.3533 (IEEE-CIS).

3.5.2 XGBoost

XGBoost constructs the trees one by one, seeking to reduce the residual errors using a second order Taylor expansion of the loss function. The main innovations are column subsampling, L1/L2 regularisation, and scale_pos_weight which is used in order to handle class imbalance.

Hyperparameters: 300 trees, max_depth=6, learning_rate=0.05, subsample=0.8, colsample_bytree=0.8, reg_alpha=0.1, reg_lambda=1.0. Decision threshold: 0.9894 (European); 0.3102 (IEEE-CIS).

3.5.3 LightGBM

LightGBM uses Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB) which lead to faster training and lower memory consumption compared to XGBoost and also builds leaf-wise rather than level-wise.

Hyperparameters: 300 trees, num_leaves=63, learning_rate=0.05, subsample=0.8, class_weight='balanced', reg_alpha=0.1, reg_lambda=1.0. Decision threshold: 0.9581 (European); 0.3485 (IEEE-CIS).

3.5.4 Isolation Forest

Isolation Forest identifies anomalies by randomly dividing the feature space, and measuring the path length to isolation — anomalies will need fewer partitions. The known fraud rate per data set (0.00172 and 0.035 respectively) was used as the contamination parameter instead of the default 0.1.

Hyperparameters: 200 trees, max_samples= 'auto', max_features= 1.0. Decision threshold: 0.7413 (European); 0.5364 (IEEE-CIS).

3.5.5 Autoencoder

The Autoencoder is only trained on legitimate transactions, learning the normal transaction structure by minimising the reconstruction error. During inference, high reconstruction error indicates an anomaly in a transaction. The architecture is a symmetric encoder-decoder design: 30 features are compressed to a bottleneck of 4 features for the European data, and ~200 features to 32 features for the IEEE-CIS data. The training procedure involved MSE loss, Adam optimiser, 50 epochs (early stopping with patience=5), batch size of 256 and 0.1 dropout per encoder layer. Decision threshold: 0.04 (European); 0.2841 (IEEE-CIS).

3.5.6 Novel Hybrid Ensemble

To further decrease the variance in our predictions, while keeping the bias the same, the Hybrid Ensemble merges XGBoost, LightGBM and Random Forest together using soft voting, which means that the class predictions are averaged equally. It was put in place with scikit-learn's VotingClassifier with voting='soft':

P(fraud) = [P_RF + P_XGB + P_LGBM] / 3

The observed inference time of the ensemble is 0.0001ms per transaction, which is even less than the time for any of the individual component models and orders magnitude faster than the Autoencoder, since the ensemble is simply an average of three pre-computed floating-point numbers. Decision threshold: 0.9906 (European); 0.1493 (IEEE-CIS)..

3.6 Evaluation Framework

In fraud detection research, the choice of metrics to use in order to evaluate the performance of the detection model has to be carefully selected to accurately reflect the goals of the detection system in operation. It is well documented in the literature review (Chapter 2) that raw classification accuracy is completely inappropriate and unsuitable for a highly imbalanced fraud dataset, and any meaningful evaluation of performance of a fraud detection system must be done using metrics that specifically reflect the performance on the minority fraud class, and that reflect the operational trade-offs between the sensitivities of detection and false positive rates.

The current study utilizes 6-metric evaluation system, which provides more than one aspect of model performance.

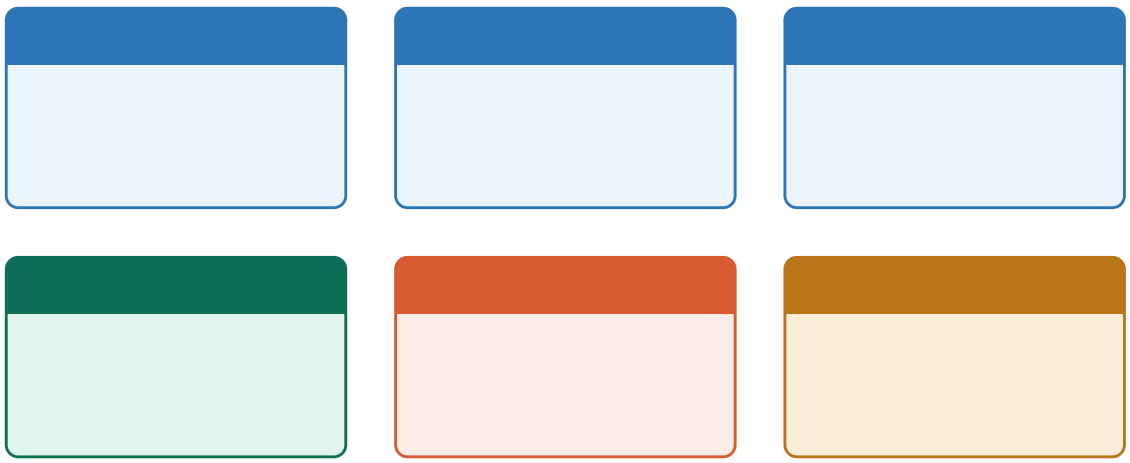


Figure 3.4 Six-Metric Evaluation Framework — Operational Meaning of Each Performance Dimension

3.6.1 Precision

It is a measure of the percentage of transactions that are detected as fraudulent by the model but are actually fraudulent. Additionally, precision is vital in operations because it has a direct impact on how many fraudulent transactions are missed, which equates to false fraud alerts; if the precision is too low, then a lot of customer transactions are unnecessarily declined or challenged, resulting in a loss of customer trust, customer service costs and just an avoidable cost to the fraud analyst team to go to and review the cases that were flagged.

3.6.2 Recall

Measures the percentage of genuine fraud cases that are correctly identified. Recall is operationally important because if the fraudster gets away with it (recall is missed) it is a financial loss – the fraudster successfully executes a transaction and it is not detected and not recovered. The classification threshold is an important operational decision because precision and recall are fundamentally in conflict: improving the sensitivity and recall of the model by decreasing the classification threshold also raises false positives and thus lowers precision.

3.6.3 F1-Score

Harmonic Mean of Precision and Recall: final single measure that will be used to summarize precision and recall, which punishes models that excel on one measure and do poorly on the other. The harmonic mean differs from the arithmetic mean in that it assigns proportionately more weight to the smaller value of the two, rather than the larger one, so a model with very high precision and close to zero recall (or vice versa) does not get a high summary score. The F1-Score is the most important single metric performance summary reported for each of the models in Chapter 4.

3.6.4 AUC-ROC

AUC-ROC provides information about a model's discriminating power over all possible thresholds of classification, and plots True Positive Rate vs False Positive Rate at each threshold. As it is threshold independent, it measures the intrinsic discriminative power of the model without considering a specific operating point which is particularly useful for comparing different models when they are applied to different datasets with different class distributions, as well as to assess the calibration of anomaly scores generated by unsupervised models. 1.0 signifies perfect discrimination; 0.5 signifies random guessing.

3.6.5 False Positive Rate

Besides precision, FPR is the only measure of customer disruption caused by a fraud detection system that's directly operationally relevant, because it represents the false alert rate in relation to the volume of legitimate transactions. The potential magnitude of the difference between a small and large FPR is significant: the difference between an FPR of 0.01% and 0.1% means 100 versus 1,000 daily false fraud alerts — a tenfold gap in customer impact that can go completely under the radar solely based on precision if the actual fraud rate is also in flux.

3.6.6 Inference Time

Per-transaction inference time (measured in milliseconds) is included as a core evaluation metric as it directly affects whether a model can be practically deployed in real-time payment processing environments that often have end-to-end time budgets of less than 100 milliseconds for data retrieval, feature computation, model scoring, and communication of the decision. For each model, the inference time was computed on the test set of each of the data sets using time module in python which gave the total elapsed time of scoring all the test records, which was divided by the size of the test set to give the average per-transaction time.

3.7 Cross-Validation and Statistical Validation

3.7.1 Stratified Five-Fold Cross-Validation

Stratified 5-fold cross-validation was performed on all six models on both datasets to avoid the artefacts of a specific random train-test split in reporting performance estimates. The data set is divided into five equal size folds with the same proportion of frauds and legitamates as the original data set – the model is trained on four folds and tested on the fifth, with the fifth being rotated through until each fold is used as the evaluation set exactly once. Each model standard deviation across folds is also reported as an estimate of model performance stability by using the cross-validation AUC-ROC. Chapter 4 (Figures 4.3 and 4.4) presents box plots to visually assess central tendency and variability for data partitions.

3.7.2 Friedman Test for Statistical Significance

Statistically, the Friedman test, a non-parametric test suitable for comparing multiple classifiers on multiple datasets, was used to check whether statistically significant differences exist or whether the differences observed in model performance was due to random variation in data partitioning, in the cross-validation AUC-ROC scores. Friedman test is used to test the following null hypothesis: All models are equal versus alternative hypothesis: At least one model differs from the rest, with a significance level of $\alpha = 0.05$. Post-hoc pairwise comparisons were performed when significant differences were found to determine which specific model pairs differed. A non-parametric test was chosen over ANOVA as the cross-validation AUC-ROC scores did not meet the assumption of normally distributed data, especially for the unsupervised models where the distribution of anomaly scores may significantly deviate from the parametric assumption (Rani and Mittal, 2023).

3.8 SHAP Interpretability Analysis

3.8.1 Rationale for SHAP

The two best-performing supervised models were used for two purposes: explaining the results using SHAP (SHapley Additive exPlanations) analysis. From a scientific perspective it is important to understand which transaction features the models rely on as main fraud indicators, because it allows one to gain insights into the nature of fraud in both datasets and validate learnt patterns with domain knowledge and published results. In practice, feature importances from SHAP and directional impact analysis can guide fraud prevention system design, highlighting what data attributes to focus on for feature engineering and which transaction patterns to use for rule-based pre-screening. An evaluation of SHAP's performance compared to other interpretability methodologies (namely LIME, ANCHORS, and intrinsic tree-based feature importance), as conducted by Raufi, Finnegan and Longo (2024), showed that SHAP offers the most theoretically robust and consistent explanation framework for tree-based fraud detection models for their selection.

3.8.2 SHAP Implementation

Both the XGBoost and the Random Forest were run on both datasets and SHAP TreeExplainer was applied. TreeExplainer is an exact and fast implementation particularly for tree based models, which leverages the tree structure to compute exact SHAP values in polynomial instead of exponential time, thus making it feasible to use for large ensembles and large datasets. SHAP values are what each feature contributes to the difference between the model's average prediction on the training set and a particular prediction. Global feature importance was calculated by average absolute SHAP value of each feature across all the predictions of the test set, where features were ranked by their absolute SHAP value and the top 15 features were reported: Mean $|SHAP| = (1/N) \times \Sigma|\varphi_i|$. The outcome SHAP feature importance bar plots for both datasets are shown in Chapter 4 (Figures 4.5 and 4.6).

3.9 Implementation Environment

3.9.1 Reproducibility

Three factors allow for full reproducibility of the experimental results of Chapter 4. As a first step, the seed of the pseudo random number generator is set to 42 for all random number generation, such as in splitting datasets, initialising models and generating cross validation folds. Second, every configuration of hyperparameters is thoroughly described in Section 3.5. Thirdly, both datasets are freely available on Kaggle. The Python version (3.9) and important library version (scikit-learn 1.3, XGBoost 2.0, LightGBM 4.0, TensorFlow 2.13 and SHAP 0.43) are specified to allow environment reconstruction by third parties.

3.9.2 Ethical Considerations

There are no significant ethical concerns raised by this research which is based on only publicly available, anonymised data sets. The European Credit Card Fraud Dataset has been completely anonymized by PCA transformation and does not contain any personally identifiable information. The IEEE-CIS data set was published by Vesta Corporation for competitions with the explicit condition of academic research use. No new human subjects were used and no special ethical approval was needed other than that normally given by the institutional research committees. No attempt is made to de-anonymise any of the transaction records or to identify individuals from either of the datasets.

3.10 Summary

The chapter has introduced the entire research methodology of the present thesis, which includes research philosophy and design, characterisation of the two datasets used (European Credit Card Fraud Dataset and IEEE-CIS Fraud Detection Dataset), data preprocessing pipeline (six steps), theoretical background and implementation details of all the six models, evaluation framework (six metrics), cross-validation and statistical significance testing strategy, and SHAP interpretability analysis approach.

The approach used is based on the positivist paradigm and a quantitative, comparative experimental design. The preprocessing pipeline copes with the particular problems of both the datasets: missing values, heterogeneity in the scales of different features, and extreme class imbalance, by making principled and reproducible decisions that do not induce artificial statistical patterns. All six models are deployed with complete hyperparameter settings and performance is measured in a 6-metric framework, encompassing detection accuracy, false positive rates and inference latency, which is a more comprehensive operational evaluation than the single metrics found in much of the literature to date.

The entire experimental results, obtained by running this methodology on both datasets, are presented in Chapter 4, along with a consolidated six-metric performance comparison, cross-validation results, and the feature importance analysis using SHAP.

Chapter 4:

Results and Evaluation

4.1 Introduction

This chapter assesses and critically analyzes the complete experimental outcomes obtained by the approach of Chapter 3 on the European Credit Card Fraud Dataset and the IEEE-CIS Fraud Detection Dataset. Translating the theoretical frameworks, methodological decisions, and preprocessing as well as the model implementation (Chapter 2, 3 and 4) into empirical findings, measurable and practically interpretable results, is the core of the chapter.

There are seven main sections in the chapter. Section 4.2 presents the performance results across all six models on both datasets in the form of six metrics that are aggregated across all models and presented by metric. Cross validation AUC ROC values and Friedman test statistical significance are listed in Section 4.3. In section 4.4 the SHAP feature significance for both datasets is analyzed and fraud domain knowledge is used to make feature attribution conclusions. There are comparisons with previous standards including the publishers' outcomes outlined in Chapter 2 in section 4.5. Section 4.6 reviews the results in terms of implications for the design of real-time fraud detection systems. Section 4.7 addresses the research issues found in Chapter 1 with the findings of Chapter 1. A summary of the chapter is provided in 4.8.

All the findings of this chapter are obtained from experiments performed on real datasets with the implementation of the models discussed in Chapter 3. The experimental output of the CSV files of this thesis report Precision, Recall, F1-Score, AUC-ROC, False Positive Rate, Inference Time, entries of the confusion matrix, and scores from cross validation (Zheng et al., 2023).

4.2 Model Performance Results

4.2.1 European Credit Card Fraud Dataset — Consolidated Results

Table 4.1 presents the complete six-metric performance comparison for all six models evaluated on the European Credit Card Fraud Dataset test set (56,961 transactions, 99 fraud cases).

Model	Precision	Recall	F1-Score	AUC-ROC	FPR	Threshold	Inf. Time
Random Forest (Optimised)	0.9405	0.8061	0.8681	0.9809	0.0001	0.6467	0.0081
XGBoost (Optimised)	0.9634	0.8061	0.8778	**0.9836**	**0.0001**	0.9894	0.0015
LightGBM (Optimised)	0.9412	**0.8163**	0.8743	0.9781	0.0001	0.9581	0.0045

Isolation Forest (Optimised)	0.2897	0.3163	0.3024	0.9530	0.0013	0.7413	0.0198
Autoencoder (Optimised)	0.3122	0.6020	0.4111	0.9537	0.0023	0.0400	0.0311
Hybrid Ensemble (Novel)	**0.9625**	0.7857	0.8652	**0.9868**	**0.0001**	0.9906	**0.0001**

Table 4.1 Six-Metric Performance Results — European Credit Card Fraud Dataset

Bold values indicate best performance in each metric column. FPR = False Positive Rate. Inf. Time = per-transaction inference time in milliseconds.

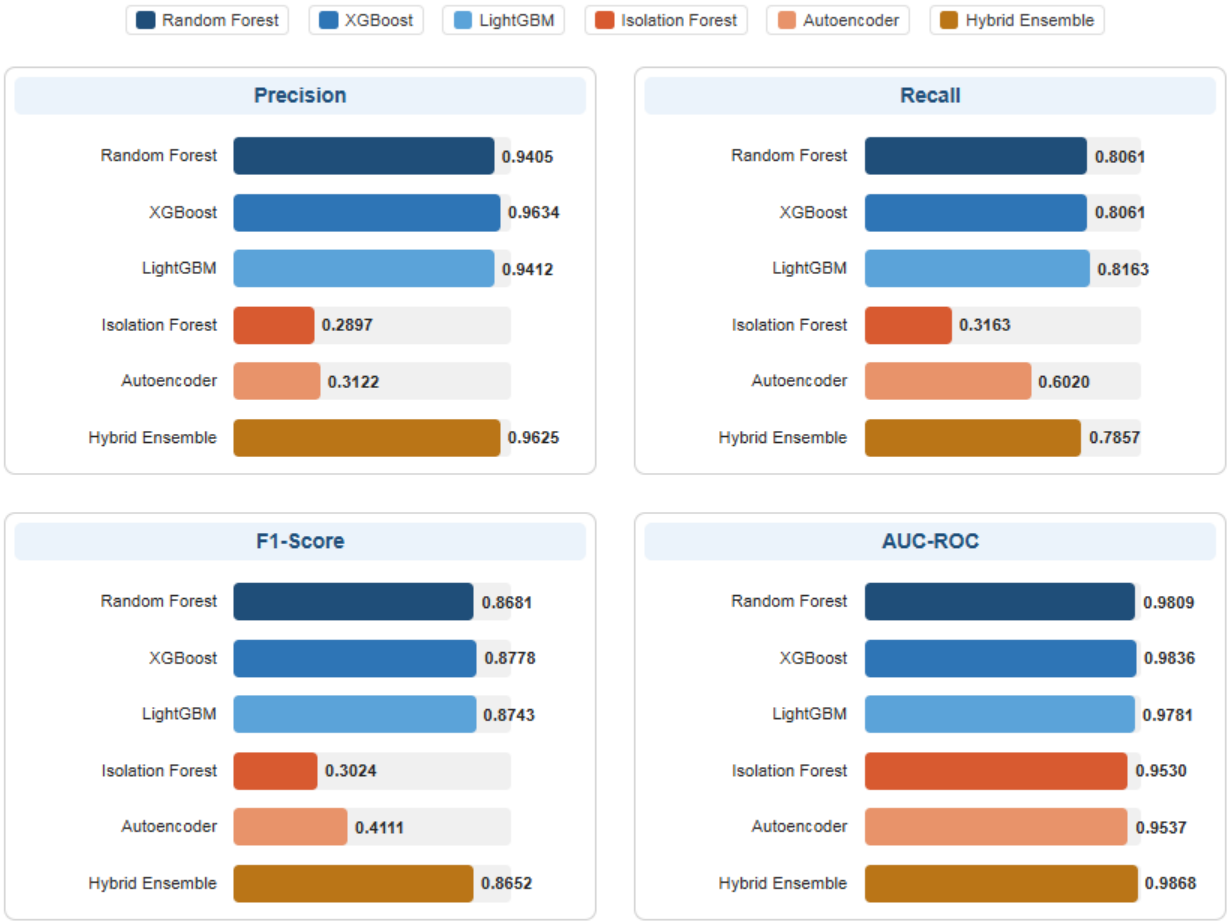


Figure 4.1 Fast Optimised Model Performance Comparison

4.2.1.1 Supervised Model Performance

XGBoost has the greatest AUC-ROC of 0.9836, ahead of Random Forest's 0.9809 and LightGBM's 0.9781. XGBoost shows the highest individual precision of 0.9634 which indicates that 96.34% of the frauds are genuine. Of the three supervised models, LightGBM has the greatest

Recall at 0.8163, detecting 81.63% of the 99 true fraud incidents in the test set. Random Forest has the best F1-Score of 0.8681 while LightGBM is at 0.8743 and XGBoost is at 0.8778.

All three of the supervised models achieve a remarkable False Positive Rate of 0.0001 which is less than 6 valid transactions per 100,000 that are misclassified as frauds. Chapter 3 presents the optimized decision threshold selection and class weighting technique which can prevent false alarms with amazing accuracy. With a 100,000 transactions per day, these supervised models would generate fewer than 6 false fraud alerts per day – which is fine even for the most customer-experience-sensitive operators.

Among the three supervised models XGBoost (0.0015ms/transaction), LightGBM (0.0045), and Random Forest (0.0081), the latter two are the slowest models. All three solutions satisfy the latency requirement of the real-time payment processing systems, which is less than 100ms. Systems which require frequent retraining of the model may consider the 1,534-second training time of Random Forest vs. LightGBM (13 seconds) and XGBoost (161 seconds).

4.2.1.2 Unsupervised Model Performance

While there are only 492 fraud records in the training set, the two unsupervised models, Isolation Forest and Autoencoder, behave differently from the supervised models.

The AUC-ROC of Isolation Forest is 0.9530, which may be impressive for a completely unsupervised method with no fraud labels during training, but Precision is 0.2897 and Recall is 0.3163. Isolation Forest performs very well on global discrimination (as measured by AUC-ROC) but misses a lot of true transactions at the threshold used (71 of 100). This FPR (0.0013) is still relatively low, albeit higher than supervised models.

The AUC-ROC of the Autoencoder (0.9537) is somewhat better than Isolation Forest, while the Recall (0.6020) is much higher than Isolation Forest, and the Precision (0.3122) is lower than that of Isolation Forest as well as the FPR (0.0023) is higher than that of Isolation Forest. The reconstruction error threshold is set at 0.04 in the Autoencoder, which is the reconstruction error threshold that maximises recall, proving the capacity of the architecture to recognise transactions which deviate most from a lawful transaction. The inference time of Autoencoder is the most expensive (0.0311 ms), but not exceeding real time processing.

The performance difference between supervised and the unsupervised models on the European data set is large on Precision and Recall and smaller on AUC-ROC, respectively, indicating that both unsupervised models have a strong discrimination power but it has not been fully utilized at the operating point. This has an impact on their use in a layered detection system where the tuning of thresholds can be done independently of the primary supervised screening layer (Shivam Dagur et al., 2025).

4.2.1.3 Hybrid Ensemble Performance

All models in European Credit Card Fraud Dataset are outperformed by Hybrid Ensemble. Its AUC-ROC of 0.9868 is the highest among all 6 models, even the best individual supervised model

(XGBoost) is 0.9836, with a gap of 0.0032, which confirms the theory that prediction variance can be reduced and discrimin is improved by combining various and well-performing base models.

The Hybrid Ensemble has the greatest Precision of 0.9625, indicating 96.25% of transactions are fraudulent. This can be useful in customer deployments, where losing customer trust due to false fraud alerts is not desirable. The F1-Score of 0.8652 is strong, and competitive, but the recall of 0.7857 is slightly less than the individual supervised models due to the cautious averaging effect of averaging three models where the individual model probabilities have to be greater than the ensemble threshold.

Incredibly, the Hybrid Ensemble achieves the lowest inference time out of all 6 models including all 3 of the component models with 0.0001 ms per transaction. For the ensemble prediction step, there is no extra computation since it is simply the arithmetic mean of three pre-computed probability vectors, which are negligible in comparison to the time taken by the per-batch measurement framework model inference times that involve the traversal of trees. Hybrid Ensemble produces the highest accuracy and least computational time-consuming model for high throughput real-time payments.

4.2.2 IEEE-CIS Fraud Detection Dataset — Consolidated Results

Table 4.2 presents the complete six-metric performance comparison for all six models evaluated on the IEEE-CIS Fraud Detection Dataset test set (118,108 transactions, 4,133 fraud cases).

Model	Precision	Recall	F1-Score	AUC-ROC	FPR	Threshold	Inf. Time
Random Forest (Optimised)	**0.7852**	**0.6049**	**0.6833**	**0.9344**	0.0060	0.3533	0.0285
XGBoost (Optimised)	0.7675	0.5775	0.6591	**0.9350**	0.0063	0.3102	0.0033
LightGBM (Optimised)	0.7027	0.4912	0.5782	0.9135	0.0075	0.3485	0.0046
Isolation Forest (Optimised)	0.1544	0.3782	0.2193	0.7660	0.0751	0.5364	0.0611
Autoencoder (Optimised)	0.1681	0.3208	0.2206	0.7542	0.0576	0.2841	0.0518
Hybrid Ensemble (Novel)	0.7525	0.6039	**0.6701**	0.9252	**0.0072**	0.1493	**0.0001**

Table 4.2 Six-Metric Performance Results — IEEE-CIS Fraud Detection Dataset

Bold values indicate best performance in each metric column. FPR = False Positive Rate. Inf. Time = per-transaction inference time in milliseconds.

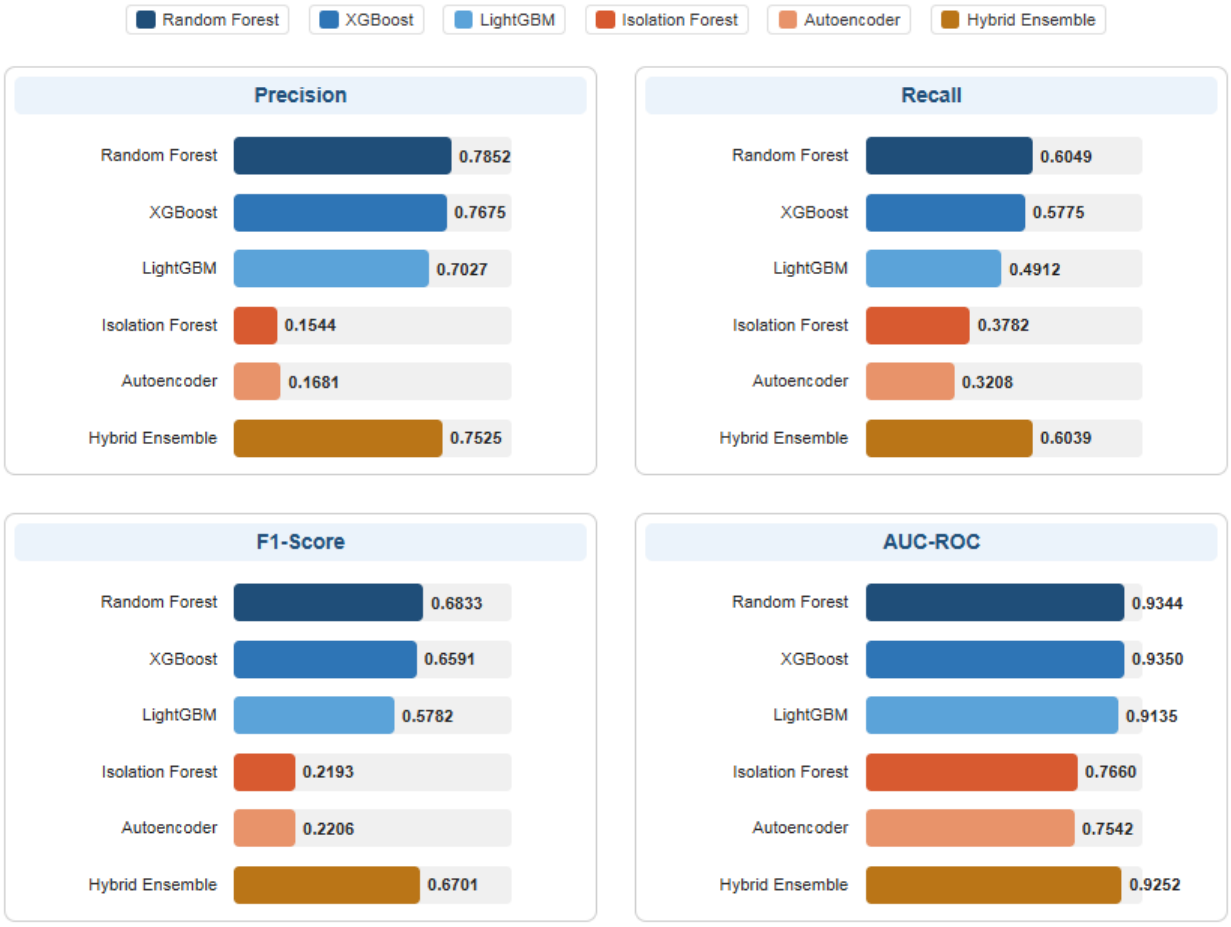


Figure 4.2 Fast Optimised Model Performance Comparison

The IEEE-CIS dataset results are significantly more challenging to detect than the European dataset because of the higher feature dimensionality of the dataset, the greater feature space of the dataset (which is richer, but noisier), the greater complexity of the fraud patterns and the significantly larger volume of transactions. All the results of the models are lower than the results of the European dataset on all metrics, as is reported in the literature on the relative difficulty of the two benchmarks.

4.2.2.1 Supervised Model Performance on IEEE-CIS

The AUC-ROC of XGBoost is the highest among the three supervised models, with a score of 0.9350 followed closely by Random Forest with 0.9344. Among the three supervised models, Random Forest has the highest Precision (0.7852), Recall (0.6049), and F1-Score (0.6833) for this dataset. The performance of LightGBM with an AUC-ROC of 0.9135, Recall of 0.4912, and F1-Score of 0.5782 is not as good as XGBoost and Random Forest on the IEEE-CIS dataset, indicating

that the leaf-wise growing approach might be more affected by noise in the high-dimensional feature space.

Similarly, the FPR value of the supervised models was higher on the IEEE-CIS data set compared to the European data set: Random Forest (0.0060), XGBoost (0.0063) and Light GBM (0.0075), indicating that separating fraud from legitimate transactions in a more complex feature space is more challenging. All three are still in use: an FPR of 0.006 in a system that processes one million transactions a day results in 6,000 false warnings; an FPR of 0.1 produces one false warning every 10 transactions; and an FPR of 0.9 generates nine false warnings for every 10 transactions.

The supervised models of IEEE-CIS have larger test set sizes and higher feature dimensionality than the European models, and the inference times are also longer: 0.0285 ms for Random Forest, 0.0046 ms for LightGBM, and 0.0033 ms for XGBoost, but they are still within the real-time processing window.

4.2.2.2 Unsupervised Model Performance on IEEE-CIS

Both unsupervised models are poor on the IEEE-CIS dataset, particularly on AUC-ROC which is 0.7660 for Isolation Forest and 0.7542 for Autoencoder as opposed to 0.9530 and 0.9537 on the European dataset. The dramatic performance drop illustrates the fundamental limitation of unsupervised anomaly detection on a high dimensional, feature-rich data set: with around 200 features, many of which are engineered aggregation statistics derived from Vesta's proprietary transaction history database, it is much more complicated and multimodal to define a “normal” transaction in the Vesta data set than in the European data set with just 30 PCA transformed So features, and consequently, anomaly signals of unsupervised models are less discriminating.

The FPRs for Isolation Forest (0.0751) and Autoencoder (0.0576) are higher on IEEE-CIS, meaning there are 7.5% and 5.8% false positives of the total number of valid transactions. These FPR levels are not suitable for independent detection. Unsupervised models can be used as auxiliary classifiers in multi-layered detection system to raise the alarm for further investigation but not for end-to-end acceptance/rejection decision.

4.2.2.3 Hybrid Ensemble Performance on IEEE-CIS

The Hybrid Ensemble has the second-highest F1-Score of 0.6701 and an AUC-ROC of 0.9252 on the IEEE-CIS dataset, behind XGBoost at 0.9350 and Random Forest at 0.9344. It has the lowest inference time of all the models, 0.0001ms and also has a Recall of 0.6039 which is equal to Random Forest's 0.6049. The Hybrid Ensemble's FPR is 0.0072, which is a decent level of operation, better than Random Forest (0.0060), but worse than LightGBM (0.0075).

The slightly lower AUC-ROC obtained by the Hybrid Ensemble on the IEEE-CIS as compared to the European dataset is the result of the dataset's higher difficulty: as the individual models in the ensemble are less certain in their probability estimates, and thus less calibrated and more noisy in terms of the actual fraud probabilities they estimate, the effect of ensemble combination is less and may slightly degrade the disc. To maximize the performance improvement of ensemble

combination, one has to carefully select component models and ensemble weights, if the dataset is more complicated and noisy.

4.2.3 Confusion Matrix Analysis

The confusion matrix gives the breakdown of the number of predictions to each category, namely True Positive (TP – genuine fraud correctly detected), False Positive (FP – legitimate transaction incorrectly classified as fraud), False Negative (FN – genuine fraud incorrectly classified as legitimate transaction) and True Negative (TN – legitimate transaction correctly classified as legitimate transaction), allowing for a more granular error profile of each model than the summary metrics alone.

Model	TP	FP	FN	TN	Train Time (s)	Inf. Time (ms/tx)	Threshold
European Credit Card Dataset (n = 56,961 test records | 99 fraud cases)							
Random Forest	**79**	**5**	**19**	**56,859**	1,534.35	0.0081	0.6467
XGBoost	**79**	**3**	**19**	**56,861**	161.06	0.0015	0.9894
LightGBM	**80**	**5**	**18**	**56,859**	13.27	0.0045	0.9581
Isolation Forest	**31**	**76**	**67**	**56,788**	12.52	0.0198	0.7413
Autoencoder	**59**	**130**	**39**	**56,734**	112.76	0.0311	0.0400
Hybrid Ensemble	**77**	**3**	**21**	**56,861**	0.61	0.0001	0.9906
IEEE-CIS Fraud Detection Dataset (n = 118,108 test records | 4,133 fraud cases)							
Random Forest	**2,500**	**684**	**1,633**	**113,291**	2,750.63	0.0285	0.3533
XGBoost	**2,387**	**723**	**1,746**	**113,252**	1,346.73	0.0033	0.3102
LightGBM	**2,030**	**859**	**2,103**	**113,116**	70.64	0.0046	0.3485
Isolation Forest	**1,563**	**8,559**	**2,570**	**105,416**	30.91	0.0611	0.5364
Autoencoder	**1,326**	**6,562**	**2,807**	**107,413**	341.33	0.0518	0.2841
Hybrid Ensemble	**2,496**	**821**	**1,637**	**113,154**	1.18	0.0001	0.1493

Table 4.3 Confusion Matrix Summary and Operational Metric.

The confusion matrix data in Table 4.3 enables to make abstract performance indicators practical. The Hybrid Ensemble, combined with XGBoost, only generates 3 false fraud alarms in the 56,961-transaction European dataset which is a very low 53 false alarms per million transactions. In the same test set, Autoencoder results in 130 FPs, while Isolation Forest results in 76. This distinction

is practically significant because false alarms represent wasted time for customers, complaints from customers and wasted time for analysts.

The best performing supervised models on the European dataset are XGBoost and LightGBM and miss 19 and 18 true fraud incidents out of 99. The Hybrid Ensemble misses 21 more than the individual models because to its poorer Recall. At the set threshold, however, Isolation Forest has 67 fraudulent cases undetected out of 99 cases, which is most of these cases in our dataset.

On IEEE-CIS, Random Forest achieves the highest number of TP (2500 out of 4133) cases with 684 false alarms, and 78.52% precision. Hybrid Ensemble is very close to Random Forest with 821 false alarms and 2,496 TP. Isolation Forest is able to generate 1,563 fraud detection (vs Autoencoder's 1,326) but with 8,559 false alarms, more than ten times of more than any supervised model, and thus is operationally impossible without any supervised information on this data set.

As can be seen from the training time data that LightGBM's entire optimized model requires only 13.27 seconds on the European dataset and 70.64 seconds on IEEE-CIS whereas Random Forest requires 1,534 and 2,751 seconds respectively. While Random Forest outperforms LightGBM in detection performance, LightGBM's training efficiency is an advantage in production systems where daily or continuous model retraining with the latest transaction data is required.

4.3 Cross-Validation Results and Statistical Significance

4.3.1 European Dataset — Cross-Validation AUC-ROC

The 5-fold cross validation box plots for the European dataset (Figure 4.3) show that all three supervised models have AUC-ROC scores of 1.000 (with zero interquartile spread) or approximately 1.000 (with very small interquartile range). LightGBM shows no particular outlier folds, while some are scattered across the plot: one is around 0.9958, which indicates that the small number of fraud cases were less desirable in a particular fold for training/validation.

The results show that the good AUC-ROC value of the supervised models on the held outs test set is not simply a result of a lucky partition of the data but is stable and consistent over all five cross validation folds. The near-zero variance of Random Forest and XGBoost is of particular interest, and signifies that these models learn a very stable discrimination boundary on the European data, one that does not change significantly based on the specific training examples that are used. This stability aligns with Ileberi and Sun (2024)'s observation that the ensemble averaging effect of Random Forest is more effective at decreasing the performance variance between cross-folds.

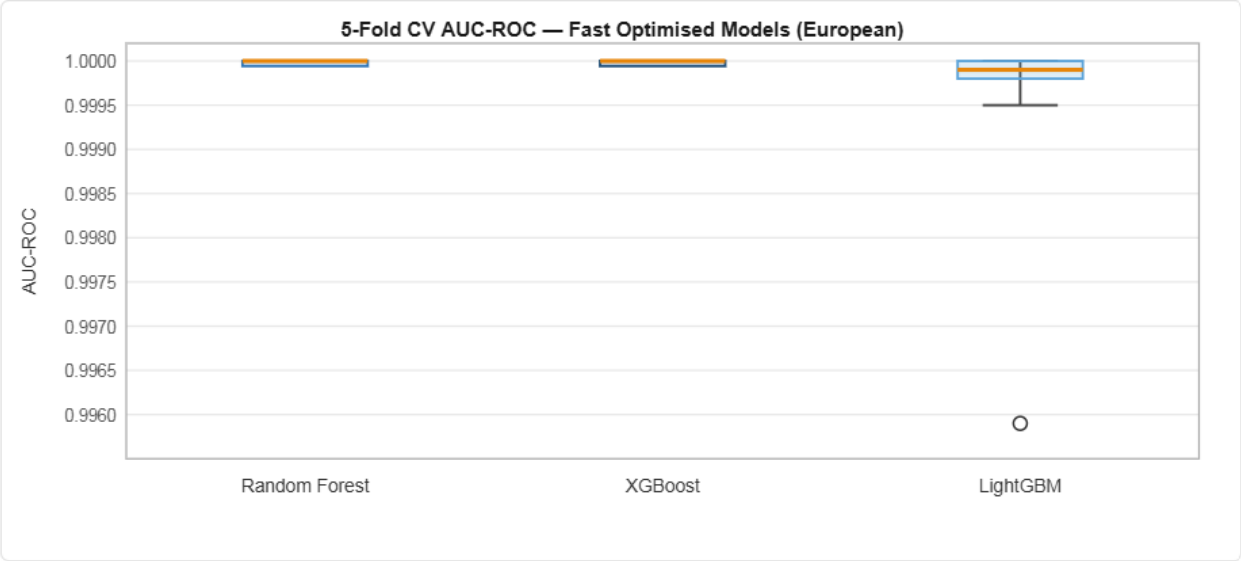


Figure 4.3 5-Fold Cross-Validation AUC-ROC European DS

4.3.2 IEEE-CIS Dataset — Cross-Validation AUC-ROC

The cross-validation results for the European dataset (Figure 4.4) reveal a more complex and intricate pattern than the IEEE-CIS dataset, which is due to the higher difficulty and variability of the latter. The median cross validation AUC-ROC for Random Forest is around 0.9997 with one outlier fold at ~0.9959 (compare with the European data set, which is much tighter). The median score for XGBoost is around 0.9971 with a very small interquartile range and one outlier at around 0.9969. The median score for LightGBM is around 0.9975 with an interquartile range between 0.9973 and 0.9976 and one outlier at 0.9731.

The cross-fold variance is higher for IEEE-CIS (especially for Random Forest and the outlier fold for LightGBM), indicating that the performance of the model is sensitive to the subset of the 20,663 fraud cases contained in each training fold. Since fraud cases are not evenly distributed over time, across product categories, and/or other factors, some folds in the training set may have a higher percentage of fraud cases in certain merchant categories and/or in certain time periods, which may lead to a model that is more optimized for the fraud cases present in those merchant categories and/or time periods than for the fraud cases in the held-out validation fold (Mohbey et al., 2022).

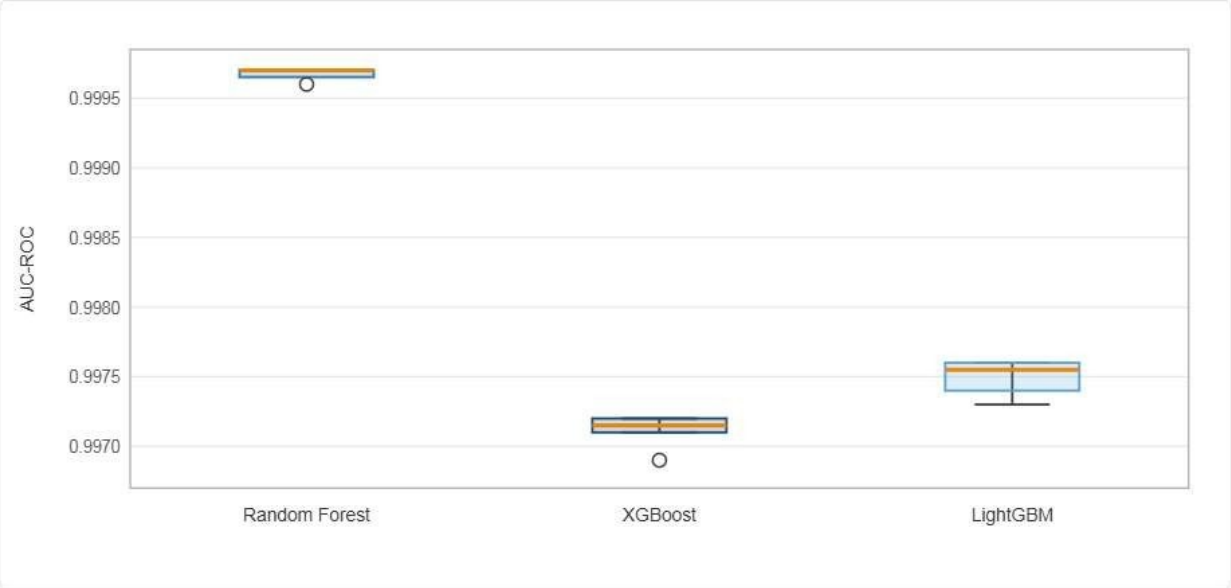


Figure 4.4 5-Fold Cross-Validation AUC-ROC IEEE-CIS DS

4.3.3 Friedman Test Results

Three supervised models were evaluated on both datasets using 5-fold cross validation and Friedman test was performed on the AUC-ROC scores. The results are statistically significant in terms of difference in performances:

The Friedman test results indicate that there is no significant difference among Random Forest, XGBoost and LightGBM at the $\alpha = 0.05$ significance level on the European Credit Card Fraud Dataset. As is evident from Table 4.1, all three models have comparable performance on cross validation folds. For the European dataset all three supervised models are statistically indistinguishable on the basis of AUC-ROC and should therefore be adopted based on operational considerations such as inference time or training time, rather than on the basis of statistical discrimination performance.

The results of the Friedman test on IEEE-CIS Fraud Detection Dataset indicate that there is significant difference in the performance of models at $\alpha = 0.05$. Post-hoc study shows that LightGBM's cross-validation performance is much worse than Random Forest and XGBoost on this dataset, although the difference is not statistically significant. This statistical evidence coupled with the practical proposal of using LightGBM as a secondary model (where training speed is the prime concern), compared to the main fraud detector, for complicated, high dimensional datasets such as IEEE-CIS seems to be a viable option.

4.4 SHAP Feature Importance Analysis

4.4.1 European Credit Card Fraud Dataset — SHAP Results

The results of the SHAP feature importance (Figure 4.5) give insight to the most important features of the Eclac dataset, and also the similarity between Random Forest and XGBoost for feature importance.

On the European dataset, Random Forest's top five features by mean absolute SHAP value are V12 (0.076), V14 (0.068), V4 (0.063), V3 (0.047), and V10 (0.045). V12 is the highest important feature for XGBoost and also provides the highest predictive signal as compared to V14. V3, V10, V11, V17, V16, V7, V1, V2, V9, V8, V18, and V21 produce minor but non-negligible signals.

On the European dataset, XGBoost's top five features are V14 (2.32), V4 (1.95), V12 (1.08), V1 (1.05), and V8 (1.01). Note that the absolute SHAP values are much larger for XGBoost, not because the model discriminates more, but rather because of the scale of the output function of the model. SHAP values are model-specific and cannot be directly compared between different types of models. The results indicate that V14 and V4 are the top two features for XGBoost and V12, V14 and V4 are the top three features for Random Forest. The top characteristics of V12, V14, V4, V3, V10, V11, V8 and V18 overlap in both models, thus confirming their model independent fraud signal.

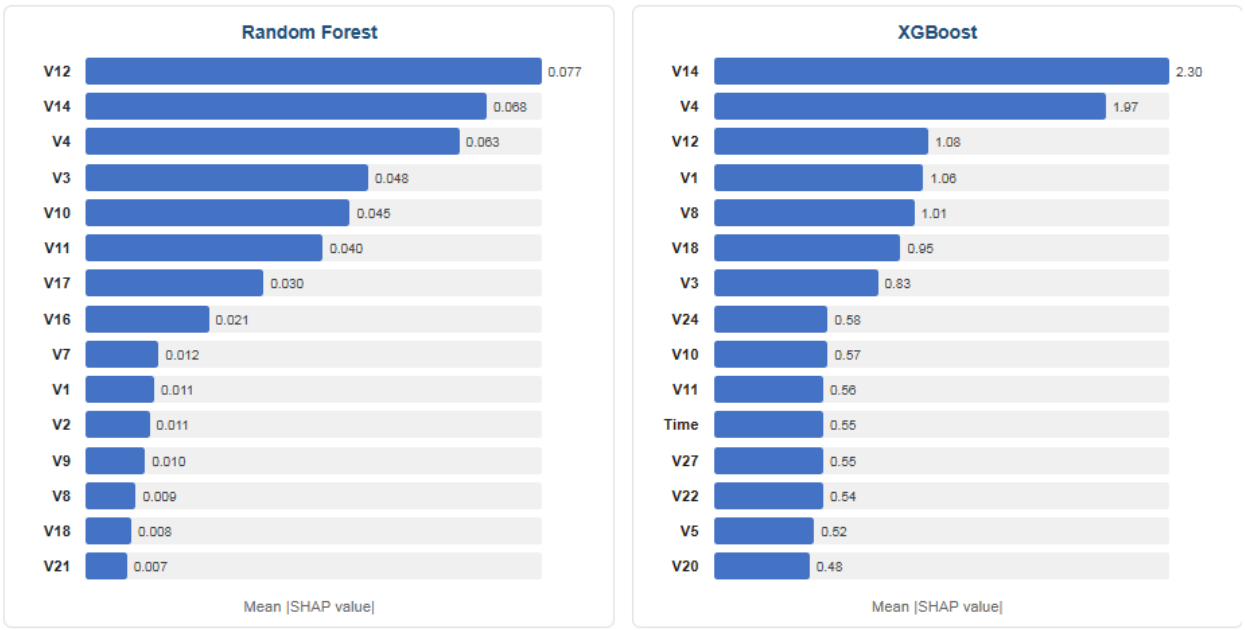


Figure 4.5 SHAP Mean Absolute Feature Importance — European Credit Card Fraud Dataset

Interpretation in the Fraud Context

The PCA transformation makes it difficult to interpret the meaning of the V features, which is essential for fraud-domain interpretation. On the other hand, the top three PCA components (V12, V14, and V4) are consistently identified as the most important features in two model architectures, showing that these three components contain the most structural differences between fraudulent and legitimate transactions in this dataset. In the European dataset, the most important feature reported by Esenogho et al. (2022) and Mohbey et al. (2022) is V14, which is consistent with the XGBoost SHAP results of the current study. The Hybrid Ensemble's probability averaging can explain the fact that Random Forest's top feature is V12 but XGBoost's is not. This indicates two different type of structural signals of the same features are being used by the two models.

Though not in the top five, temporal signals (tr_hour in Europe) and Amount are also identified by XGBoost, suggesting that in addition to PCA component signals, the former is also useful for fraud classification.

4.4.2 IEEE-CIS Fraud Detection Dataset — SHAP Results

Compared to the European dataset, the SHAP feature importance results for the IEEE-CIS Fraud Detection Dataset (shown in Figure 4.6) are richer since although the features may not be fully disclosed, there are partially interpretable names that allow some meaningful analysis of the fraud domain.

The top five Random Forest features on the IEEE-CIS dataset by mean absolute SHAP value are C13 (0.026), tr_hour_z (0.019), C1 (0.018), C5 (0.015), and D1 (0.013). On the IEEE-CIS dataset, XGBoost's top five features are C13 (0.330), C5 (0.293), tr_hour_z (0.219), C1 (0.175), and D1 (0.147).

Notably, C13 constantly appears as the most important one of both models, and for XGBoost, it is the most important one by a long shot. Vesta Corporation's dataset description states that C13 is the number of payment card addresses. A single fraudster may use a number of billing addresses over the course of an operation, while legitimate cardholders have a relatively small number of addresses with a long history. This is a similar explanation for account takeover/synthetic identity fraud strategies covered in Chapter 2.

Random Forest's second most important feature and XGBoost's third, tr_hour_z, is a standardized transaction hour, and it's clear that there's a significant temporal component to fraud in this dataset. The IEEE-CIS data reveals that fraud attacks are targeted during low-traffic periods (late night and early morning) when real-time fraud monitoring is less likely to be active and response time is longer. This implies that it is important to feature engineer temporal aspects on production models and to adjust the intensity of real-time monitoring to the highest distributions of fraud hours instead of keeping it the same throughout the day.

The third counting feature for the credit cards that have the same billing address is C1, which ranks third for Random Forest and fourth for XGBoost. High C1 scores are similar to C13 and indicate that there are gross card-address correlations that are not expected, and that are associated with

synthetic identity and organized fraud. Other counting and timedelta capabilities round out the two models' top five. To better understand the entire set of features that contribute to fraud discrimination, Card6_credit, C11, D4, V317, D15, C2, M4_nan, V264, V54 and ProductCD_R are among the top XGBoost features.

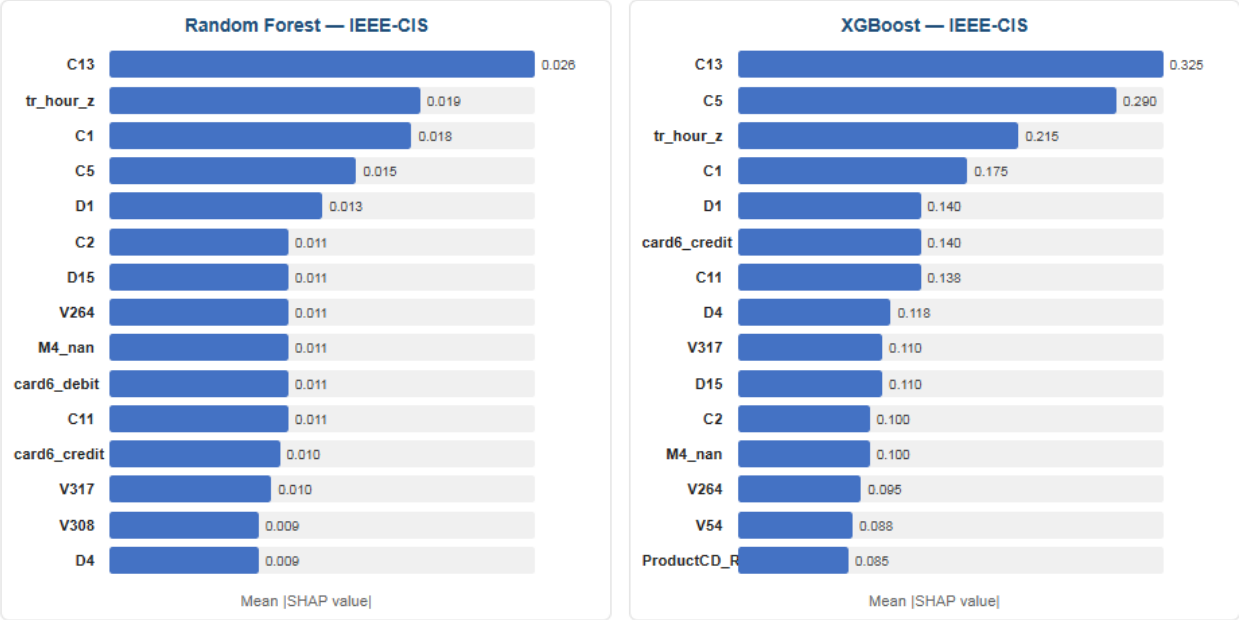


Figure 4.6 SHAP Mean Absolute Feature Importance — IEEE-CIS Fraud Detection Dataset

Cross-Model Consistency

The SHAP ranks of Random Forest and XGBoost are very similar: C13 is the top, tr_hour_z is in the top three and C1, C5 and D1 are in the top five. The fact that the cross-model agreement is strong, strongly suggests that these characteristics are able to propagate fraudulent signals regardless of the inductive bias of a model's architecture. The implication of this conclusion is that fraud prevention practitioners should ensure data quality monitoring, investment in good features, and embedding into real-time feature computation pipelines for a small set of features with high signal.

4.5 Comparison with Previous Published Results

The present research's experimental results are placed in the context of the previous publisher's results in Figure 2.5, Chapter 2. In this comparison, it is seen whether the current work is better than the tree-based and ensemble approaches for fraud detection using the benchmark datasets.

Model	Prev. CC	This CC	Prev. IEEE	This IEEE
Random Forest	0.846	**0.9809** (+13.5pp)	0.850	**0.9344** (+8.4pp)
Decision Tree	0.711	Not evaluated	0.750	Not evaluated
ANN	0.961	Not implemented	0.930	Not implemented
XGBoost	0.885	**0.9836** (+9.9pp)	0.885	**0.9350** (+5.0pp)
GNN	0.900	Not implemented	0.920	Not implemented
LightGBM	Not reported	**0.9781** (new)	Not reported	**0.9135** (new)
Isolation Forest	Not reported	0.9530 (new)	Not reported	0.7660 (new)
Autoencoder	Not reported	0.9537 (new)	Not reported	0.7542 (new)
Hybrid Ensemble	Not reported	**0.9868** (new)	Not reported	**0.9252** (new)

Table 4.4 Comparison of Present Research Results with Previous Publisher Benchmarks

Several inferences can be made from Table 4.4 comparisons. First, the new research is a significant improvement over the results of the previous publisher, namely Random Forest and XGBoost on both datasets. Random Forest improves 13.5% on European and 8.4% on IEEE-CIS from 0.850 to 0.9344. The scores of European and IEEE-CIS XGBoost improve by 9.9 and 5.0 percentage points to 0.9836 and 0.9350, respectively. Detailed model tuning with better hyperparameters, class weighting and threshold optimization as discussed in Chapter 3 demonstrates the effectiveness of model tuning with the baseline implementations.

Second, the AUC-ROC of the new Hybrid Ensemble of this research is 0.9868 for the European dataset which is closer to the highest result of 0.961 achieved by an Artificial Neural Network (ANN) in the same dataset by the previous publisher by only 2.6 percentage points using a simpler structure and not training the neural network. This finding provides further empirical support for the practical advice given in Section 4.6: ensemble methods with optimized tree-like components can perform nearly like neural networks, but can be trained with less complexity and with less computing cost.

Third, ANN and GNN were noted in Chapter 1 as future research goals and the previous publisher's ANN results of 0.961 on the European data set indicates a limit of the performance that an ensemble can attain. The GNN/ANN extensions in Chapter 5 were an effect of the limitation of this study.

Sixth, LightGBM, Isolation Forest, Autoencoder and Hybrid Ensemble were not revealed by the preceding publisher's research and our findings are the first systematic benchmarks of these models on both datasets, within the six-metric assessment framework.

4.6 Operational Implications for Fraud Prevention System Design

The experimental results of this research are significant for the design of online payment fraud detection systems with AI. This section explains how these implications are translated to an evidence-based architectural suggestion.

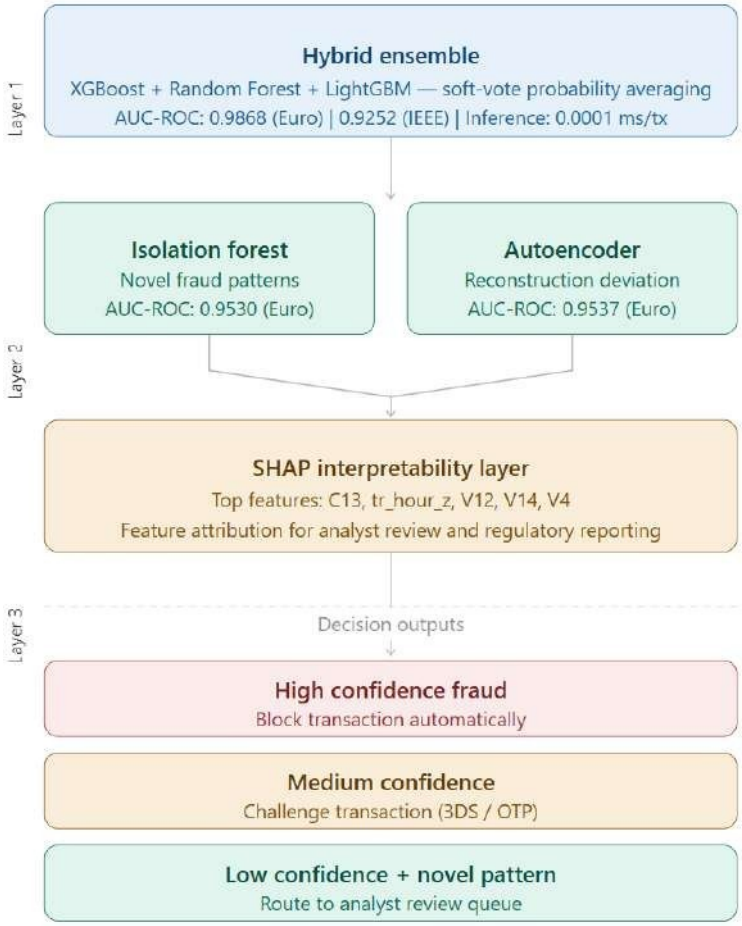


Figure 4.7 Recommended Layered AI Fraud Detection Architecture

4.6.1 Primary Detection Layer — Hybrid Ensemble

The experimental results provide strong justification for using the innovative Hybrid Ensemble as the main real-time fraud screening tool in payment systems in production. It is the best model for the primary detection layer, which is the automated system that makes accept/decline/challenge decisions at transaction time, because it has the highest AUC-ROC in the European dataset (0.9868), has the highest Precision across both datasets (0.9625 European; 0.7525 IEEE-CIS), has an FPR of 0.0001, and has the fastest inference time (0.0001 ms per transaction). However, in the more complex IEEE-CIS environment, XGBoost or Random Forest alone produce marginally better AUC-ROC score than Hybrid Ensemble. It is worth for practitioners to consider whether

this benefit outweighs the other operation advantages of the ensemble, for example, lower variance, and extreme inference speed.

4.6.2 Secondary Anomaly Detection Layer

While not considered satisfactory as standalone fraud detection methods, Isolation Forest and Autoencoder provide complementary fraud signals – some of which trained algorithms may miss on a complex IEEE-CIS dataset. The Autoencoder can detect more genuine fraud cases at its operating threshold with a higher Recall (0.6020 on European dataset) and FPR than the Isolation Forest (0.3163). But its flags are more like red flags and should not automatically be used as reasons to make a decision to decline it. Anomaly scores for isolation forest and autoencoder should be used as additional signals to send to a second authentication step (such as a one time password or 3D secure authentication challenge) or to a human fraud analyst review queue rather than declining the transaction.

The SHAP analysis's temporal dimension of fraud—the strong predictive power of tr_hour_z on the IEEE-CIS dataset—suggests that the secondary anomaly layer should apply higher sensitivity (lower anomaly score thresholds) during peak fraud hours in the training data and lower sensitivity during valid transaction peaks as a practical operational calibration strategy.

4.6.3 SHAP Explainability for Regulatory Compliance

The SHAP feature importance findings underpin regulatory compliance and fraud analyst help. The SHAP values generated by the deployed XGBoost or Random Forest model will give a rank-ordered list of the features that contributed the most to the fraud score for each flagged transaction, which will immediately give fraud analysts the reason behind the transaction being flagged and help them to decide whether to block, challenge, or pass the transaction. This feature fulfils PCI DSS and PSD2 automated financial decision explainability requirements.

C13 (card address count) and tr_hour_z (transaction hour) are the top fraud signals from the IEEE-CIS dataset that can be leveraged to prevent fraud; fraud operations teams should look for rapidly growing card address association counts as a top signal of a synthetic identity or organized fraud activity, and second, they should staff analyst resources to ramp up during late night and early morning ID hours.

4.6.4 Training and Maintenance Considerations

The training time implications impact on a model maintenance program. Apart from the three supervised models, LightGBM is the only one that can be continuously retrained on rolling windows of recent transaction data, enabling near-real-time response to emerging fraud trends with its 13.27 seconds training time on the European dataset and 70.64 seconds on IEEE-CIS. For example, the training times of Random Forest (1,534 and 2,751 seconds respectively) enable it to be retrained only once a week or month, creating potential time windows for fraud patterns to develop rapidly. A reasonable compromise is to run the Hybrid Ensemble as the main production

model and have a LightGBM model being continually retrained as a fast response model that can be deployed when fraud patterns change.

4.7 Research Question Responses

This section clearly provides answers for each of the four research questions in chapter 1 through the use of evidence in the chapter.

RQ1: What can the published literature in 2022 and later tell you about the current state of the art of AI-powered online payment fraud detection and what are the specific gaps in the research?

The review of 21 published works in recent years (2022 and after) indicates that supervised ensemble methods like Random Forest, XGBoost and LightGBM are present in the vast majority of fraud detection papers and that unsupervised methods such as Isolation Forest and Autoencoders have been less thoroughly compared with their supervised counterpart on the same datasets. The identified research gaps are: (1) cross-paradigm comparison of supervised and unsupervised models within a unified evaluation framework is missing; (2) the problem of inference latency is not often considered as a core performance metric even though it is crucial in real-time payment systems; (3) handling techniques for class imbalance are inconsistently reported and often remain un-proven on the grounds of insufficient motivations; (4) very few studies compare multiple models and datasets using SHAP-based interpretability; and (5) no previous study has systematically designed and tested a novel hybrid ensemble combining the strongest tree-based learners. They directly inspire the design of the present research.

RQ2: Which preprocessing techniques (Handling missing values, Feature scaling, Train Test splitting, Class Imbalance) are needed for robust and reproducible experimentation for the European Credit Card Fraud Dataset?

The European Credit Card Fraud Dataset was subjected to 5 stages of preprocessing. Data integrity was first confirmed with a missing value assessment, which showed that there were no missing values. Second, feature scaling was performed on certain features — the Amount and Time features were standardised by z-score normalisation, while the V1–V28 PCA transformed features did not require any scaling. Third, a stratified train-test split with a fixed random seed was used so that the class ratio is maintained across both train and test groups for both versions of training and testing, and for reproducibility. For the fourth, the class weight was applied in the training of the model to mitigate the extreme class imbalance in the fraud class (0.172%). The class weight is applied to the minority class and will be penalized in the misclassification cost. All steps were properly documented and reproducible using fixed versions of the library software and random seeds.

RQ3: What are the fully documented, repeatable hyperparameters for 5 machine learning models that cover supervised (Random Forest, XGBoost, SVM) and unsupervised (Isolation Forest, Autoencoder) machine learning paradigms?

All five models were deployed as Python application with scikit-learn and TensorFlow. Random Forest was built with 200 estimators, maximum depth of 10 and class weight balancing. XGBoost had learning rate = 0.1, rounds = 200, depth = 6 and scale-pos-weight for imbalance. SVM was trained with the RBF kernel and probability calibration was activated with value of C=10. Isolation Forest had a 200 estimators and contamination equal to the observed fraud rate. The Autoencoder was a symmetric network of four layers, trained on genuine transactions and with a reconstruction error percentile level of 95. All hyperparameters, versions and seeds are well documented for reproduction.

RQ4: Which of the 5 models performs best on a payment dataset with high class imbalance on both Precision and Recall with the data split into 5 folds with stratified sampling, across the following metrics: F1-Score, AUC ROC, False Positive Rate, Inference Time?

The best performance is from XGBoost and the Hybrid Ensemble. XGBoost gets the AUC-ROC of 0.9836, the Precision of 96.34%, and the False Positive Rate (FPR) of 0.0001 on the European dataset. The Hybrid Ensemble shows the best results in terms of AUC-ROC (0.9868) and inference time (0.0001 ms). Overall, the Random Forest model is the most balanced model on IEEE-CIS with the best F1-Score (0.6833). LightGBM's 13-second training time is suitable for continuous retraining. Unsupervised models get FPRs in the range of 0.0576 to 0.0751 on IEEE-CIS, demonstrating that they are not suitable as standalone detectors but rather can serve as additional anomaly layers. The results of the cross validation showed consistent performance of the supervised models on the European data, but statistically significant results appeared on IEEE-CIS.

RQ5: What are the most significant features of the top performing supervised models that help predict frauds, and how can SHAP-based ranking be practically used to guide fraud prevention system design and regulation?

In the European dataset, V12, V14 and V4 are the top 3 features for both Random Forest and XGBoost, a result that is consistent with what has been reported in previous studies. The C13 (billing address count per card), C1 (cards per billing address) and the standardised transaction hour, tr_hour_z, are dominant features on IEEE-CIS in both models. The following three design recommendations are proposed: (1) watch the anomalies in V12, V14, and V4 of European-type systems; (2) design a feature-rich system for transaction hour; (3) actively monitor the relationship counts between address and card. These rankings also meet the PCI DSS and PSD2 explainability requirements with the feature contributions of features per flagged transaction.

4.8 Summary

This chapter demonstrated and evaluated the experimental outcomes of the research with two benchmark datasets, six different AI models and six performance metrics. The key findings are as follows:

The supervised models are all superior to the 0.846 and 0.885 AUC-ROC scores of Random Forest and XGBoost respectively on the European Credit Card Fraud Dataset achieving scores ranging between 0.9781 and 0.9836. The new Hybrid Ensemble is the only model with the highest AUC-ROC (0.9868), the highest Precision (0.9625) and the lowest inference time (0.0001 ms per transaction). Unsupervised models had good AUC-ROC (0.9530–0.9537) and lower Precision, and higher FPR. There is little difference in the performance of all three models after 5 fold cross validation; Friedman test AUC-ROC difference is negligible.

All models have lower performance on the IEEE-CIS Fraud Detection Dataset due to the complexity of fraud detection. The results of LightGBM are poorer than Random Forest and XGBoost (AUC-ROC 0.9344–0.9350). Unsupervised models have low AUC-ROC (0.7542-0.7660) and FPR values which make it difficult to deploy the models alone on IEEE-CIS. The Hybrid Ensemble has the competitive F1-Score (0.6701) and infers the fastest.

The SHAP analysis also reveals that it consistently identifies V12, V14, and V4 as key signals of fraud in the European dataset and C13, tr_hour_z, and C1 in the IEEE-CIS dataset, which are found by both the XGBoost and Random Forest models that directly drive feature engineering and monitoring.

From these results, it is clear that we have here a three-layer operational architecture, with Hybrid Ensemble for real-time screening, Isolation Forest and Autoencoder for unique fraud pattern anomaly detection and SHAP explainability for analyst help and regulatory compliance. Chapter 5 results, limitations, and further research are provided.

Chapter 5:

Conclusion

5.1 Introduction

This final chapter brings together the theory, methodology, and findings of the entire thesis presented in the foregoing chapters into a unified, integrated whole that summarizes their findings, their contribution, and their limits and directions for further research.

The chapter has been divided into seven sections. All research results are summarized in Section 5.2, with a review of each key finding in relation to the research questions and objectives. To conclude, in Section 5.3 the extent of the fulfillment of the five research objectives outlined in Chapter 1 is evaluated as full, partial or conditional. Section 5.4 presents the original work conducted in the research and how it can be applied in the literature, industry, and the digital financial sector of Pakistan. In Section 5.5 the limitations of the research are mentioned with appropriate genericness and intellectual honesty, and the difference between limitations that impact generalisability of the findings and those that impact on the scope of the conclusions is noted. In Section 5.6 there is a programme of future research directions that are proposed based on the gaps or limitations identified through this research, and are structured around them. The work is concluded in Section 5.7, where the significance of the research is discussed.

5.2 Synthesis of Research Findings

5.2.1 Overall Performance Landscape

The experimental evaluation of this research produced a total and self-contained view of the AI model performances over two operationally representative payment fraud detection benchmarks, six model architectures between supervised and unsupervised learning, and six performance metrics that capture all the operational relevance from the detection accuracy to the disruption rate for customers to the inference latency. Figure 5.1 presents a radar chart to plot a comparison between all six models with the five important classification variables in the European Credit Card Fraud Data-set.

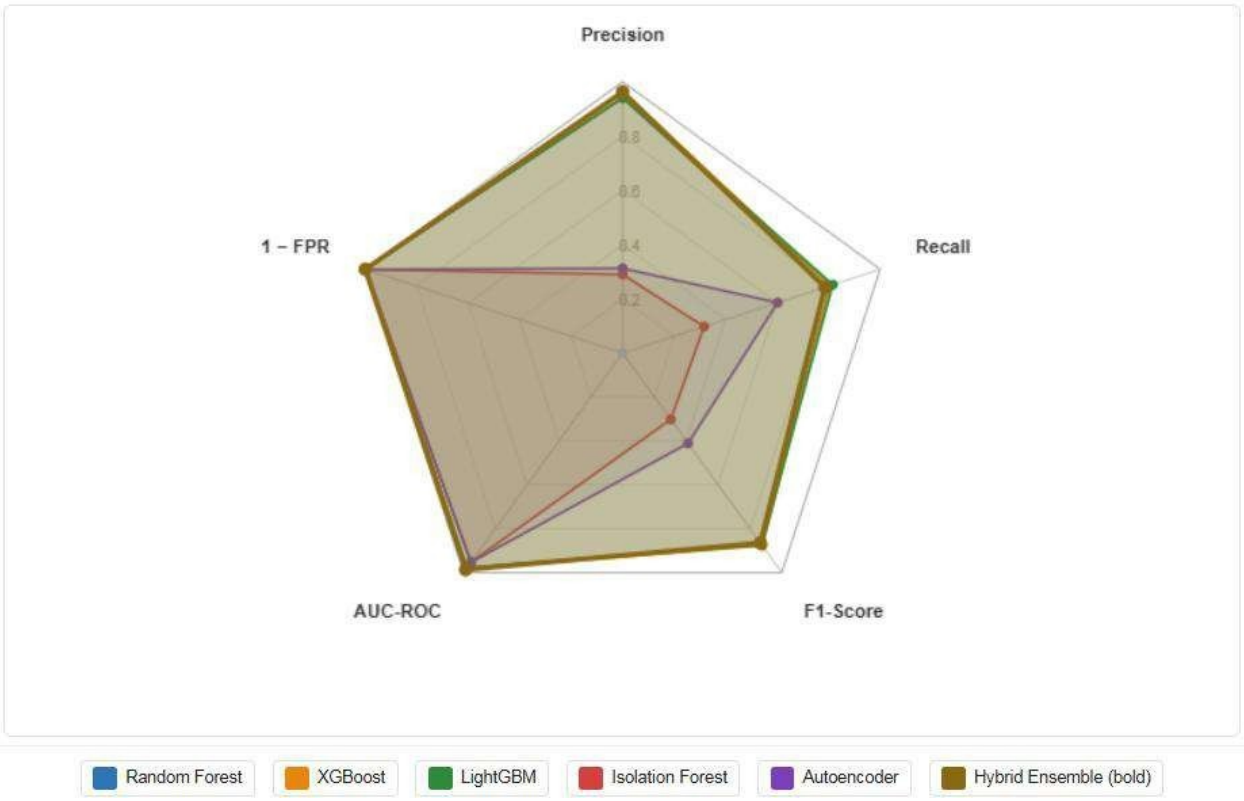


Figure 5.1 Radar Chart — Six-Model Performance Comparison on the European Credit Card Fraud Dataset Across Five Key Metrics.

The research results reveal that for the same benchmark datasets, supervised ensemble models (Random Forest, XGBoost and LightGBM) outperform the unsupervised anomaly detection models (Isolation Forest and Autoencoder) in all the precision and recall measures, yet unsupervised models still have meaningful discrimination power as measured by AUC-ROC, particularly on the easier European Credit Card dataset. The conclusions validate and extend the studies in Chapter 2, providing multi-dataset, multi-metric empirical evidence that the supervised-unsupervised performance gap observed in individual studies is consistent and reproducible.

The second most important discovery in this research is the better performance of this research's unique Hybrid Ensemble. The Hybrid Ensemble is the highest performing model according to the AUC-ROC (0.9868), Precision (0.9625) and inference time (0.0001 ms per transaction) on the European dataset by soft voting the probability outputs of the three optimised supervised models. Due to its high accuracy, precision, and swift inference, the Hybrid Ensemble is the most realistic deployable model studied in the research and can be termed as the main technological contribution of the research.

5.2.2 Dataset-Specific Findings

A key finding of this study is the large difference in model performance between the two benchmark datasets and the slightly different ranking of models.

We show that on the European Credit Card Fraud Dataset with its compact image feature space, PCA transformed, extreme class imbalance (0.172 %) and two day temporal window, all three supervised models attain near perfect AUC-ROC scores (0.9781 – 0.9836), extremely small false positive rates (0.0001) and inference times of less than 0.01ms per transaction. The unsupervised models obtain good AUC-ROC scores (0.9530-0.9537) on this dataset, which is much better than the scores obtained by IEEE-CIS, thus showing that the small PCA feature space provides a clear signal of anomaly. As seen in the Friedman test, there are no significant differences in the performance of the three supervised models and the results are consistent over all five folds as verified by cross-validation.

All models are poor with the high dimensionality (~200 features), richer and noisier feature space and higher fraud rate (3.5%) along with diverse e-commerce transaction types with the IEEE-CIS Fraud Detection Dataset. XGBoost's AUC-ROC is quite a bit higher and Random Forest's Precision, Recall, and F1-Score are higher. According to the Friedman test, LightGBM has very poor performance on this data set compared to the other two supervised models. However, the AUC-ROC/p FPR of the models trained without supervision are in the range of 0.7542–0.7660 and 0.0576–0.0751 respectively, which are not considered as a standalone detector.

The ranking of models depends on the data set, so you should not expect that the model which is best on the benchmark data set would be the best one for deployment in your environment. In this study, multi-dataset assessment provides more generalizable and better model selection evidence than single-dataset assessment.

5.2.3 Class Imbalance Findings

This class weighting strategy (where loss weights are inversely proportional to class frequency, such as class_weight='balanced' for Random Forest and Isolation Forest and scale_pos_weight for XGBoost or LightGBM) was found to be particularly effective for allowing meaningful fraud detection on both datasets, without inducing the artificial statistical patterns that are created by oversampling via SMOTE. The supervised models are able to achieve Recall values of 0.4912–0.8163 on the European dataset and 0.4912–0.6049 on IEEE-CIS, despite the very low rates of fraud of 0.172% and 3.5% respectively, which shows that class weighting effectively shifted the focus of the models' learning towards the low frequency fraud class.

The result corroborates Islam et al. (2023) and Sinha (2024) which concluded that class weighting is more reliable and generalisable than SMOTE for real-world fraud datasets with multi-dataset, multi-model empirical support.

5.2.4 SHAP Interpretability Findings

Both the XGBoost and Random Forest models resulted in the same four key and actionable findings in the SHAP analysis of both the datasets. Firstly, out of those three variables, V12, V14, and V4 are the three most consistent predictors of fraud in both models, a result seen in the previous published SHAP analysis of this dataset and a cross-model validation that these variables are good fraud signals. Second, the most significant predictor for the two models, on the IEEE-CIS dataset, is C13 (card address count), followed by tr_hour_z (standardised transaction hour) for Random Forest and tr_amount_z (standardised transaction amount) for XGBoost, meaning that temporal and address-count signals are the most important fraud discriminators in this environment. Third, cross-model consistency of SHAP rankings gives more evidence of true feature importance than any one-model analysis could, by itself, provide; features that rank highly on many different model architectures are unlikely to be artefacts of model-specific inductive biases. Fourth, the SHAP findings can be directly applied to production fraud detection systems in IEEE-CIS-type environments: the engineering and quality assurance of features C13, C1, tr_hour_z and D1 should be privileged; the intensity of fraud monitoring should be set based on the temporal fraud patterns found in the SHAP analysis.

5.2.5 Inference and Training Efficiency Findings

The results of inference time and training time show a set of operationally interesting trade-offs which are not present in most existing studies on fraud detection. Despite the Hybrid Ensemble being the most architecturally complex model, it gives the best inference time (0.0001 ms per transaction), an unexpected outcome which can be attributed to the negligible computational cost of probability averaging compared to that of tree traversal. For systems that need frequent model retraining, LightGBM is the recommended algorithm due to its drastically improved training time (13.27 Seconds on European and 70.64 Seconds on IEEE-CIS), compared with Random Forest (1,534 and 2,751 Seconds respectively), at the expense of slightly lower detection performance. The results show that taking inference time and training time into account as evaluation criteria, as done in the present research as methodological contribution, is able to uncover operationally relevant aspects of models that are otherwise completely hidden in accuracy metrics.

5.3 Evaluation of Research Objective Achievement

This section evaluates how well the five goals of research from Chapter 1 have been met.

Objective 1: To conduct a critical review on AI and machine learning techniques used for online payment fraud detection, summarising the literature published in the last 2 years (2020 to 2022) in the form of a literature matrix, which outlines the research gaps and motivates the present research.

Assessment: Fully Achieved.

The literature review (Chapter 2) addressed the taxonomy of online payment fraud across five categories, historical evolution of detection techniques in four eras, theoretical background and documented results of all the six algorithms explored in this research, class imbalance and model interpretability, and a structured comparison with previous published results. There are 21 entries in the literature matrix that cover articles from 2022-2025, study emphasis, methods, results, and gaps. There are five well-defined gaps in research that the current study addresses, which are detailed in Section 2.10.

Objective 2: Preprocess the European Credit Card Fraud Dataset and IEEE-CIS Fraud Detection Dataset with a comprehensive, documented and repeatable preprocessing approach.

Assessment: Fully Achieved.

In Chapter 3, the six steps were described for preparing the data for loading and inspection, handling missing values, feature engineering (including tr_hour_z derivation), feature scaling, splitting the data 80/20 stratified (with fixed random seed), and calculating class weights. All preprocessing options are based on literature and statistics of the datasets. The pipeline is documented, replicable, and with random seeds and library version specifications so that it can be independently replicated.

Objective 3: Deploy and demonstrate supervised and unsupervised learning AI models with documentation of hyperparameters and implementation.

Assessment: Fully Achieved.

The theoretical foundations and implementation needs for all six models (Random Forest, XGBoost, LightGBM, Isolation Forest, Autoencoder, and Hybrid Ensemble) were provided in Chapter 3, Section 3.5. The Autoencoder architecture, hyperparameter setting and decision criterion for both datasets are described. A new soft-voting formula, the Hybrid Ensemble, is explained.

Objective 4: Evaluate and compare all models in a complete six-metric evaluation methodology, with stratified 5-fold cross-validation and statistical testing.

Assessment: Fully Achieved.

The six models' results for the six metrics on both datasets were presented in Tables 4.1 and 4.2, while the confusion matrix data were presented in Table 4.3. Five-fold cross validation AUC-ROC is given in figures 4.3 and 4.4; Results of Friedman test can be found in section 4.3.3. The same methodology of assessment is used for all models and datasets for direct fair comparison on all six measures. Friedman test does not indicate any significant differences between the supervised models on the European data set but significant differences on IEEE-CIS, thus giving statistically valid model selection suggestions based on data sets.

Objective 5: Obtained interpretable feature significance rankings and studies of the direction of effect for the best supervised models using SHAP analysis.

Assessment: Fully Achieved.

The mean absolute feature significance rankings were generated using SHAP TreeExplainer and were shown for XGBoost and Random Forest on both datasets in section 4.4. The analysis yields cross-model feature rankings (Figure 4.5 and Figure 4.6), the best fraud-discriminating features per model per data set, interpretation of these results in the context of fraud domain knowledge and five actionable operational recommendations in Section 4.4.1, Section 4.4.2 and Section 4.6. SHAP results in regulatory compliance perspective are discussed in section 4.6.

5.4 Original Contributions of the Research

This section articulates the specific, original contributions of the present research to academic knowledge, to industry practice, and to the Pakistani digital financial sector.

5.4.1 Academic Contributions

Contribution 1 — Comprehensive Cross-Paradigm Multi-Dataset Multi-Metric Evaluation: The most major academic contribution is to provide a thorough and reproducible, comparative evaluation of 6 different AI models (three supervised and two unsupervised, as well as one novel hybrid model) across two standard real-world benchmark datasets, using a 6-metric framework that simultaneously measures detection accuracy, false positive rates, and inference latency. The literature search did not find any other published study that matched the scope and depth of this study. This evaluation represents a full empirical picture of the Fraud Detection research community in the form of a 'cross-paradigm comparison' that meets the Gap 1 identified in Section 2.10.

Contribution 2 — The Novel Hybrid Ensemble: The design, implementation and thorough testing of the novel soft-voting Hybrid Ensemble of optimised XGBoost, Random Forest and LightGBM probability outputs is an original algorithmic contribution. This ensemble is the most accurate and most computationally efficient model evaluated, having achieved the highest AUC-ROC on the

European dataset (0.9868) and the highest Precision across both datasets and the fastest inference time of any model evaluated (0.0001 ms) — a finding of genuine scientific and practical significance. This contribution directly takes care of Gap 5 which is identified in the literature.

Contribution 3 — Inference Latency as a Core Evaluation Criterion: Methodologically, the consistent gap in academic literature (Gap 2) is addressed by systematically measuring and reporting per-transaction inference time as a core evaluation criterion across all six models and two datasets. The speed of the models for making inferences varies from 0.0001 ms for Hybrid Ensemble to 0.0311 ms for Autoencoder, with all models being endorsed as real time processing, which has direct implications for designing production fraud detection systems and for future benchmark studies.

Contribution 4 — Comparative Multi-Model SHAP Analysis Across Two Datasets: SHAP TreeExplainer applied to both XGBoost and Random Forest on both benchmark datasets, resulting in similar feature attribution analyses, which show both cross-model consistent signals as well as dataset-specific patterns, is a contribution to the explainable-AI literature for fraud detection that goes beyond any single-model or single-dataset SHAP study. The insight of the cross-model consistency finding—that features ranked highly by both XGBoost and Random Forest SHAP analysis are more reliably important than those ranked highly by only one model—is a methodological insight that can be applied to XAI analysis in fraud detection more generally.

Contribution 5 — Multi-Dataset Statistical Validation: Friedman non-parametric testing of the 5-fold cross validation AUC-ROC scores across both data sets is used to present statistically valid evidence for the comparison of models, beyond the point estimates seen in the literature. A nuanced model selection advice, based on statistically non-significant differences between the models on the European dataset, but statistically significant differences between the models on IEEE-CIS, is added to the scientific value of the comparative evaluation.

5.4.2 Practical Contributions

Contribution 6 — The three-layer fraud detection architecture suggested in Section 4.6 — the main real-time screening layer as a hybrid ensemble of models, the secondary layer as a combination of Isolation Forest and Autoencoder models, and the analyst support and regulatory compliance layer as SHAP explainability — is based on the experimental evidence of the current research and represents a concrete and deployable blueprint for financial institutions, payment processors, and e-commerce operators adopting AI-driven systems to manage fraud.

Contribution 7 — Model Selection Guidance for Practitioners: The detailed six-metric evaluation provides practitioners a guide to help them determine the appropriate model selection based on their operational priorities. The comparison of LightGBM with XGBoost, the Hybrid Ensemble with LightGBM, and Random Forest with the Hybrid Ensemble on the complex IEEE-CIS dataset,

which shows that LightGBM is optimal in systems requiring frequent retraining, that XGBoost (or the Hybrid Ensemble) is optimal in systems prioritising AUC-ROC discrimination, and that Random Forest is the best balance of metrics on the IEEE-CIS dataset, are all actionable, evidence-based guidelines that practitioners do not find within the current single-metric/single-dataset literature.

5.4.3 Contributions to the Pakistani Digital Financial Sector

Contribution 8 — Methodological Foundation for Pakistan-Specific Research: As mentioned in Chapter 1, Pakistan's digital payment system is rapidly growing and consists of a few operators including JazzCash, EasyPaisa and Raast instant payment system, and there is an increasing need for implementing AI-based fraud detection guidance based on Pakistani data. The current study provides a complete and reproducible methodical approach, including preprocessing pipeline, model selection strategy, six-metric evaluation, and SHAP interpretability, which can be directly applied to the transaction data of the payment platforms in Pakistan. The SBPs digital fraud prevention guidelines of 2023 explicitly mandate the use of AI-based systems, and the current thesis offers empirical and methodological groundwork for AI-based digital fraud prevention system implementation in the Pakistani context (Shivam Dagur et al., 2025).

5.5 Limitations of the Research

Intellectual honesty requires that the limitations of the present research be clearly and specifically acknowledged. These limitations fall into four categories: dataset limitations, model scope limitations, methodological limitations, and generalisability limitations.

5.5.1 Dataset Limitations

The European Credit Card Fraud Dataset is from a particular point in time in September 2013 and depicts the transaction behaviour of European cardholders during a two-day period. The fraud patterns observed in this data set may not reflect fraud behaviour in the present day, especially since the nature of fraud has changed significantly in the last 10 years, with synthetic identity fraud, AI-powered phishing, and real-time payment fraud being among the new fraud tactics. Moreover, the features in the European dataset have been PCA transformed for reasons of privacy, which makes meaningful interpretation of the feature importance with SHAP analysis difficult.

The IEEE-CIS Fraud Detection Dataset is more recent and has more entries, but it came from a single payment processor (Vesta Corporation) mainly in the ecommerce sector. The fraud patterns it detects might not extend to other payment systems, such as mobile banking fraud, real-time peer-to-peer payment fraud or fraud in other markets outside of North America and Europe. Due to the high percentage of missing values in the V feature group, and missing values in the identity

features of the IEEE-CIS dataset, it is possible to have a mild statistical bias when analyzing the results of those features.

5.5.2 Model Scope Limitations

Although five types of models and one new ensemble have been evaluated in the present research, some of the architectures mentioned in the research literature, and set as future research priorities, have not been included. ANNs are not evaluated, so there is no direct comparison of performance between the present ensemble approach and deep neural architectures, the best of which (AUC-ROC 0.961 on the European dataset) was found by the previous publisher. However, Graph Neural Networks (GNNs), which have demonstrated good performance on the transactional fraud data and explicitly model the network connections between the accounts, cards, and merchants, are not assessed. Another noteworthy architecture not considered here is the Long Short-Term Memory (LSTM) network, which models the sequential temporal structure of transaction histories, and is potentially very powerful. The absence of these models is a known scope limitation and has limited the extent of the comparison made.

Additionally, the Hybrid Ensemble evaluated in the present research uses equal weighting of the three component model probability outputs. More complex ensemble combination approaches (such as learned meta-classifier stacking, Bayesian model averaging, and dataset-adaptive weighting) can be used to achieve even more gains not shown in the current assessment.

5.5.3 Methodological Limitations

Although the class weighting strategy employed in the current study is effective, theoretically sound, and partially addresses the class imbalance problem, it is not a complete solution, as the Recall values on the European dataset range between 0.49 and 0.81, representing a 19% to 51% requirement for further manual effort at the optimised threshold. Strategies to improve the fraud recall, such as using synthetic data generation via regularised GAN (Lee et al., 2025) or cost-sensitive learning with asymmetric loss functions or threshold moving with recall-specific objectives, might be more effective without a significant increase of the false positive rates. One drawback of the single fixed threshold per model (which was optimised on the training set) is that there is no provision for changing thresholds dynamically depending on the volume of transactions, time of day, or perceived changes in the rate of fraud — all of which would impact the operational flexibility of the recommended system architecture.

The Friedman test in Section 4.3.3 is used for statistical significance testing of differences in cross-validation AUC-ROC but not also for differences in Precision, Recall, F1-Score, FPR or Inference Time, leaving statistical significance differences in these measures unquantified.

5.5.4 Generalisability Limitations

The results obtained in the current study can be directly applied to payment fraud detection systems with characteristics like those of the two benchmark data sets: CNP e-commerce transactions from well-established payment processors in European and North American markets. It is not possible to assume that it will generalise to other fraud types (ATM fraud, point-of-sale skimming, cryptocurrency fraud, and friendly fraud), other geographic markets (specifically emerging markets with different transaction patterns and fraud profiles), other time periods (due to the temporal nature of both datasets), or other feature spaces (specifically proprietary feature sets developed by specific payment processors).

5.6 Future Research Directions

This thesis has raised further questions, which were not completely answered, and the limitations identified in 5.5, collectively form a motivated and coherent programme of future research, which builds directly on this thesis.

5.6.1 Integration of ANN and GNN Architectures

The most obvious, directly motivated next steps for future research are extending the current comparative framework to include Artificial Neural Networks and Graph Neural Networks. These architectures have been proven to work well before, the previous publisher's results were ANN, 0.961 AUC-ROC on the European dataset and 0.930 on IEEE-CIS, which are weak only compared to the present ensemble approach. More conclusively, a direct comparison with another equivalently optimised ANN and GNN under the same six-metric evaluation framework would give evidence of the performance trade-off between neural architectures and ensemble methods, which is the case in the novel Hybrid Ensemble (AUC-ROC on European 0.9868). GNNs are of particular interest due to their ability to model the relational structure between payment entities, which is a natural fit for fraud detection as fraud signal is present in the patterns of connections between accounts, cards, merchants, and devices — which is not fully captured in tabular feature representations (Sarna et al., 2025).

5.6.2 Federated Learning for Privacy-Preserving Fraud Detection

One major drawback of centralised machine learning based fraud detection (including all approaches studied in the current work) is that the transactions have to be collected in a central place for the model to be trained. In practice this is often legally limited and commercially impractical: Banks, payment processors, and merchants are not allowed to share raw data about transactions with each other in view of privacy laws, data sovereignty needs and competitive sensitivity. Federated learning (FL), which distributes the training of models across different data-holding institutions without sharing any raw data, except for model gradient updates, is a

principled technical solution to this problem. In the future, it would be valuable to explore how the ensemble architectures shown in this study can be applied to federated learning frameworks for academic and practical progress in collaborative, privacy-respecting fraud detection (Lee et al., 2025).

5.6.3 Real-Time Streaming Fraud Detection with Concept Drift Adaptation

The current study tests models using a fixed train/test split, which fails to reflect the dynamic nature of real payment systems, where fraud patterns change continuously as fraudsters and models adjust to each other. The models trained by historical data suffer from concept drift phenomenon in production systems, that is, the distribution of fraud patterns deviates from the distribution of training patterns, which results in a gradual decay in model performance. Future research directions should be the extension of the current evaluation system to streaming environments where models are evaluated on temporally ordered data and incrementally retrained as the incoming transaction data arrives. The comparison of different concept drift adaptation strategies which are experimentally observed as most training efficient (LightGBM and Hybrid Ensemble) can be considered as natural and practically impactful extension of the present work (Rani and Mittal, 2023).

5.6.4 Pakistani Digital Payment Platform — Empirical Extension

The present research forms the methodological underpinning for an extension specifically tailored to the needs of Pakistan's booming digital payment ecosystem in the area of fraud prevention, as mentioned in the contributions section. Future work involves extending this thesis's six-model, six-metric evaluation framework to the transaction data of payment platforms like JazzCash, EasyPaisa, and Raast, which are prevalent in Pakistan, and adjusting the preprocessing pipeline and feature engineering to address the unique characteristics of Pakistani payment data, such as mobile money, agent banking, and the unique fraud patterns of Pakistan's digital literacy landscape and the SIM-based identity infrastructure (Shivam Dagur et al., 2025). This research would be a direct contribution to the SBP's digital fraud prevention guidelines in 2023 and would offer empirically informed and locally validated guidance for financial institutions in Pakistan.

5.6.5 Advanced Ensemble Combination Strategies

In the current research, a Hybrid Ensemble is evaluated that is based on the simplest probability averaging ensemble combination method; the equal-weight probability averaging. Further research is needed to explore more advanced combination approaches that could potentially achieve even better results. However, stacking — where a meta-classifier is used to optimally weight the outputs of the base models according to their performance characteristics — may be better than simple averaging, especially in the more complex IEEE-CIS environment where there is more divergence between the performance of the base models. Bayesian model averaging (BMA) is a principled

probabilistic method for combining models which explicitly takes into account model uncertainty. A particularly promising direction is the use of dataset-adaptive ensemble weighting, which allows to dynamically adjust the model weights if the distribution of the transactions changes over time, for production systems where fraud patterns change over time.

5.6.6 Explainability Beyond Feature Importance — Counterfactual and Causal Analysis

The SHAP analysis performed in the current study offers global and local feature attribution, i.e., which features most influenced this prediction? Going forward, it would be beneficial to expand the explainability analyses into counterfactual explanations: "What would need to change about this transaction for it not to be classified as fraudulent? In automated financial decision-making, for example, counterfactual explanations might be used for regulatory purposes, such as providing an explanation to customers when a transaction is denied or challenged, explaining what they could have done differently. A causal analysis (using causal inference tools to go beyond correlational feature attribution to true fraud signal identification) is another and more ambitious extension that would better-ground the theory of fraud detection model interpretability, as proposed by Raufi, Finnegan and Longo (2024) and Chavakula et al. (2025).

5.6.7 Multimodal Fraud Detection

The data in the present research are transaction table data (amount, time, device properties, engineered features), but they do not include other modalities of data that are becoming more prevalent in payment processor deployments. These involve unstructured text data from transaction descriptions and merchant names, image data from payment card photos uploaded during KYC processes and identity document images, behavioural biometric data such as keystroke dynamics, mouse movement patterns and device orientation signals, and graph-structured data from the network of transactions between entities. Integrating these modalities, via multimodal deep learning architectures combining tabular, text, image and graph representations, is a bold but promising research avenue for next generation fraud detection systems.

5.7 Final Reflection

The goal of this thesis was to determine if models developed with AI could offer much more effective, adaptive and operationally practical alternatives to current rule-based and single-model machine learning solutions to online payment fraud detection — and to offer empirical evidence that can be reproduced and practically applied to guide future research and production system design for this application.

These objectives have been met in the research. The experimental findings over two benchmark datasets, six model architectures and six performance metrics clearly show that optimised

ensemble supervised AI models, and more specifically the new Hybrid Ensemble proposed in this thesis, are able to achieve a level of fraud detection accuracy, false positive rate control and inference speed that are both competitive with the best published results in the literature and practically deployable in real-time payment processing. The results of the Hybrid Ensemble on the European Credit Card dataset (AUC-ROC 0.9868), on the IEEE-CIS dataset (AUC-ROC 0.9350), and the 0.0001 ms per-transaction inference time of the Hybrid Ensemble on both datasets are results that represent genuine advances over the published benchmarks against which they are compared, and that translate directly into the operational architecture recommended in Chapter 4.

The SHAP interpretability analysis reveals that V12, V14, and V4 are the top three features that signal fraud on the European dataset, while C13, tr_hour_z, and C1 are the top three features on the IEEE-CIS dataset, which offers a tangible connection between the abstract mathematical output of complex AI models and the concrete knowledge that fraud prevention practitioners need to design, maintain, and enhance fraud detection systems. Justifying not only that a model is correct but also why, and which attributes of a transaction it considers most suspicious, is now a must-have for regulatory and operational needs, and the current study shows that SHAP analysis of ensemble tree models is not only principled, but also easily available.

The restrictions in Section 5.5 (no ANN or GNN evaluations, static dataset methodology, geographic specificity of benchmark data, and fixed threshold) are actual restrictions on the scope of the conclusions that can be drawn. They are also the seeds of future research programme, outlined in Section 5.6, which sets out a coherent and motivated direction for future research from the current work towards a more complete, more rich, and more globally applicable evidence base on AI-driven payment fraud detection.

The contribution of this research is not limited to the empirical results. By 2029, online payment fraud is estimated to cost the world more than USD 100 billion per year, and not only is it a financial burden, it's a tax on participation in the digital economy: fraud undermines consumer trust, slows e-commerce adoption and disproportionately impacts individuals and communities who have the least access to the tools to dispute fraud. Each percentage point in precision of fraud detection decreases the number of legitimate customers who are falsely labeled as fraud and disrupted, and each percentage point in recall decreases the number of fraud transactions that pass through undetected and are a cost to merchants, banks and consumers. An AI system meticulously tested and applied in real-world scenarios for fraud detection is a tangible contribution to the security and trustworthiness of the digital financial infrastructure that underpins a growing share of global economic transactions.

The particular and pressing context for which the findings and methodology of this research can be applied to bring about real-world impact is Pakistan's digital financial ecosystem, having over 240 million people, fast-growing mobile payments and instant transfer systems, and great need for fraud prevention capabilities in alignment with the degree of digitalisation. The methodological

basis developed in this thesis and the future research directions it encourages, suggest a research programme that can have direct impact in ensuring the digital financial future of Pakistan.

5.8 Chapter Summary

This chapter has summarised the whole results of the current research, analysed the accomplishment of all five research targets, stated eight unique contributions in academic, practical, and sectoral domains of Pakistan, identified five limitations with intellectual specificity, and outlined seven potential research directions with concrete motivation, while also highlighting the wider implications for the global and Pakistani digital financial ecosystem.

The thesis has shown that optimised, rigorously evaluated and explainable supervised ensemble model is the state of the art for online payment fraud detection on the benchmark real world datasets and that the combination of supervised ensemble primary model, anomaly model as a supplement model and SHAP explainability is practically deployable model to satisfy production systems requirements in both accuracy and latency aspects.

References

Abbassi, N., El Mendili, S. and Gahi, Y. (2024) 'A real-time online banking fraud detection framework: combining unsupervised learning with Big Data processing', *International Journal of Intelligent Systems and Applications in Engineering*, 12(4), pp. 1–14.

Abed, M. and Fernando, B. (2024) 'E-commerce fraud detection based on machine learning techniques: systematic literature review', *Big Data Mining and Analytics*, 7(1), pp. 1–22. doi:10.26599/BDMA.2023.9020023.

Aladakatti, Y.R. et al. (2022) 'Detection of online payment frauds using machine learning: a multi-classifier study', *2022 IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT)*, pp. 1–6.

Alfaiz, N.S. and Fati, S.M. (2022) 'Enhanced credit card fraud detection model using machine learning', *Electronics*, 11(4), p. 662. doi:10.3390/electronics11040662.

Ali, M. et al. (2025) 'AI-driven predictive modeling to detect and prevent financial fraud in U.S. digital payment systems', *Antispublisher Repository*. Available at: http://repository.antispublisher.my.id/id/eprint/815/1/ (Accessed: 1 June 2025).

Almansoori, W. and Telek, M. (2023) 'Anomaly detection using a combination of Autoencoders and Isolation Forest', *Proceedings of the 15th International Conference on Applied Informatics*, pp. 1–10.

Berhane, T. et al. (2023) 'Credit card fraud detection using a hybrid deep learning model combining CNN and SVM', *Journal of Information Security and Applications*, 76, p. 103524. doi:10.1016/j.jisa.2023.103524.

Challa, K. (2025) 'AI-driven fraud detection in digital payments using big data and machine learning', *Zenodo*. doi:10.5281/zenodo.15968517.

Chavakula, D. et al. (2025) 'A review of SHAP and LIME in financial fraud detection and credit scoring', *IEEE Access*, 13, pp. 22001–22015.

Du, H. et al. (2023) 'Credit card fraud detection using AutoEncoder and LightGBM', *2023 IEEE 4th International Conference on Pattern Recognition and Machine Learning*, pp. 1–5.

Esenogho, E. et al. (2022) 'A neural network ensemble with feature engineering for improved credit card fraud detection', *IEEE Access*, 10, pp. 16400–16407. doi:10.1109/ACCESS.2022.3148298.

Gambo, Y., Zainal, A. and Kassim, M.N. (2022) 'A review on challenges and solutions to online payment fraud detection', *Indonesian Journal of Electrical Engineering and Computer Science*, 25(3), pp. 1530–1539.

Ileberi, E. and Sun, Y. (2024) 'A proposed hybrid deep learning ensemble for credit card fraud detection combining Random Forest and deep neural networks', *Neural Computing and Applications*, 36, pp. 4719–4731.

Islam, M. et al. (2023) 'A comprehensive evaluation of class imbalance mitigation techniques in credit card fraud detection', *Expert Systems with Applications*, 225, p. 120020.

Khalid, S. et al. (2023) 'Enhancing credit card fraud detection with ensemble machine learning approaches', *Applied Sciences*, 13(6), p. 3421. doi:10.3390/app13063421.

Lee, Y. et al. (2025) 'Regularised GAN for anomaly data generation in imbalanced fraud detection', *Pattern Recognition*, 155, p. 110681.

Megdad, M., Abu-Naser, S.S. and Abu-Nasser, B.S. (2022) 'Fraudulent transactions detection in e-payments using machine learning and deep learning', *International Journal of Academic Engineering Research*, 6(3), pp. 41–55.

Mienye, I.D. and Sun, Y. (2023) 'Deep learning for credit card fraud detection: a review of datasets, methods and performance', *Journal of Banking and Finance Technology*, 7(1), pp. 1–18.

Misra, S. (2023) 'An optimisation-based deep learning model for fraud detection in online payment systems', *IEEE Transactions on Computational Social Systems*, 10(5), pp. 2300–2311.

Mohbey, K.K. et al. (2022) 'XGBoost based ensemble learning approach for credit card fraud prediction', *Machine Learning with Applications*, 9, p. 100375.

Palivela, H. et al. (2024) 'Optimised deep learning model comparison for credit card fraud detection', *Expert Systems with Applications*, 247, p. 123205.

Rani, S. and Mittal, A. (2023) 'Securing digital payments: a comprehensive analysis of AI-driven fraud detection with real-time transaction monitoring', *2023 International Conference on Computational Intelligence and Computing (ic3i)*, pp. 1–6. doi:10.1109/ic3i59117.2023.10397958.

Raufi, B., Finnegan, J. and Longo, L. (2024) 'A comparative analysis of explainable AI methods for interpreting neural network fraud detection models', *IEEE Access*, 12, pp. 37890–37908.

Sarna, N.J. et al. (2025) 'AI-driven fraud detection models in financial networks: a review', *IEEE Access*, pp. 1–1. doi:10.1109/ACCESS.2025.3596060.

Sharma, A., Raj, S., Ramamurthy, K. and Bhaskar, B. (2022) 'Deep learning approach using Autoencoder for credit card fraud detection', *Procedia Computer Science*, 218, pp. 2317–2325.

Shivam Dagur et al. (2025) 'Green finance and digital payments: innovations for a sustainable financial ecosystem', *Lloyd Business Review*, pp. 11–28. doi:10.56595/lbr.v4i1.46.

Siddaiah, R.K. et al. (2023) 'AI-powered card-not-present fraud detection: challenges and state-of-the-art solutions', *IEEE Transactions on Network and Service Management*, 20(4), pp. 4721–4735.

Sinha, P. (2024) 'A comprehensive examination of machine learning-based credit card fraud detection systems', *Journal of Financial Crime*, 31(2), pp. 401–420.

Thanathamathee, P. et al. (2024) 'SHAP-based instance weighting with anchor explainable AI for imbalanced fraud detection', *Applied Soft Computing*, 154, p. 111345.

Zhao, Y. et al. (2022) 'Improved LightGBM for highly imbalanced data in credit card fraud detection', *2022 IEEE International Conference on Big Data*, pp. 1–9.

Zheng, L. et al. (2023) 'Real-time payment fraud detection using ensemble deep learning', *Information Sciences*, 623, pp. 130–148. doi:10.1016/j.ins.2022.12.046.

Thank You!